Remarks on Mixed Holomorphic-Topological Strings

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Abstract

This paper studies a local constrained system. The formal holomorphic normal surface has coordinates z_1, z_2 and symplectic form $dz_1 \wedge dz_2$. On a stack of N probes these coordinates become matrices ϕ_1, ϕ_2. The open ghost-zero theory is the Dirac first-order system

$$\int_{\mathbb{R}} \operatorname{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Here A is the multiplier for the first-class constraint $[\phi_1, \phi_2] = 0$, and the BV/Koszul variable $\psi = A^*$ records it by $Q\psi = [\phi_1, \phi_2]$. This is the open local model for a mixed holomorphic-topological BV sector on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $\mathbb{R} \times p$.

After removing constant Hamiltonians and passing to the degreewise stable connected trace sector, the classical open Hamiltonian algebra is

$$\mathbf{S}\big(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]\big), \qquad \bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1,$$

where $\bar{A}_{\mathrm{desc}}$ is represented by the nonconstant primitive one-antifield Koszul cycles. The closed Hamiltonian sector is cotangent BF theory for the reduced Tate Lie algebra

$$\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \qquad \mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[1],$$

with bracket induced by the usual Poisson bracket after projection modulo constants. Its local observables are the continuous Chevalley-Eilenberg cochains $C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g})$. The reduced derived Poisson center of the degree-zero open algebra is computed explicitly:

$$C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) \cong \big(PV_{\mathrm{cont}}(\widehat{\mathbf{S}}(\mathfrak{h})), [\pi_\partial, -]\big),$$

with $c_I \mapsto \partial/\partial O_I$ and $u_I \mapsto O_I$. Thus the boundary Hamiltonian function is the image of the shifted cotangent CE coordinate, not of the nonclosed CE coordinate dual to a Hamiltonian mode.

The Koszul dual of the closed CE algebra is the completed enveloping algebra $\widehat{U}(\mathfrak{g})$. Its PBW special fibre identifies degree by degree with the connected stable open Hamiltonian trace sector as a filtered graded-commutative product. The polynomial descendants are PBW special-fibre labels; they do not realize the closed coadjoint deformation-level action. Principal parts provide the correct coadjoint algebraic model. Boundary evaluation on Hamiltonian modes

$$f(z_1, z_2) \longmapsto \overline{\operatorname{Tr} f(\phi_1, \phi_2)}$$

identifies the degree-zero generators and intertwines the closed Hamiltonian Poisson bracket with the connected open trace bracket. This degree-zero Hamiltonian current map has an interval-wise ordinary central-operation representative after the de Rham jet contraction and satisfies the P_0 bracket identity on stable connected local cohomology. The full unreduced BV/factorization-center lift remains a problem: compact-support functoriality before contraction, P_0 center homotopies, principal-part boundary currents, and descent must all be constructed.

The formal deformation-quantization target is the $\hbar$-adic Weyl/Moyal deformation. The first-order bracket is proved, and the all-order Weyl/Moyal local coefficients are computed algebraically. At finite N, Weyl-ordered traces descend through quantum Hamiltonian reduction; the Harish-Chandra radial-parts theorem proves the reduced Moyal commutator identity for the Weyl/Capelli-renormalized generators. The stable connected quantum quotient and the factorization-level quantum theorem remain open. The scalar Hamiltonian/center-of-mass term and its Capelli quantum form are treated as a named obstruction. The reduced open-line midpoint model gives the first and third Weyl graph weights, but the paper does not construct the Costello closed-open Feynman weights. The remaining analytic problem is to identify the Costello boundary graph expansion, in this local Hamiltonian model, with the Weyl/Moyal coefficients, including anomaly control, stable connected trace normalization, and the first and third graph normalizations.

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0.1 Introduction

0.1.1 Overview

The local problem is a constrained canonical system. A brane sitting at a point of a holomorphic symplectic surface has two normal coordinates. On a stack of N probes those coordinates become two matrices ϕ_1, ϕ_2, with symplectic potential $\mathrm{Tr}(\phi_1\, d\phi_2)$. Change of Chan-Paton basis imposes the first-class moment-map constraint

$$[\phi_1, \phi_2] = 0,$$

and the ghost-zero open action is the Dirac first-order action

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

The BV/Koszul variable $\psi = A^*$ records the constraint by $Q\psi = [\phi_1, \phi_2]$.

A closed Hamiltonian field $f(z_1, z_2)$ is measured by the brane by substituting the matrix-valued normal coordinates:

$$z_i \mapsto \phi_i, \qquad f \longmapsto \mathrm{Tr}\, f(\phi_1, \phi_2).$$

The degree-zero theorem proves that this measurement is a moment-map shadow: after removing the scalar Hamiltonian and stabilizing in N, the boundary bracket is the Poisson bracket of the normal symplectic plane.

The closed Hamiltonian BF sector is the shifted cotangent Lie algebra $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ of Hamiltonian vector fields on the formal disk. Its local observables are continuous CE cochains. The reduced derived Poisson center of the stable degree-zero open Hamiltonian algebra is computed below and is canonically $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$. Separately, the PBW special fibre of the Koszul dual $\widehat{U}(\mathfrak{g})$ identifies with the connected stable open Hamiltonian trace sector as a filtered graded-commutative product. These are different statements.

Remark 0.1.1. This introduction records the physical source of the local theorem. The mathematical assertions are the formal local statements proved below.

Let X be a holomorphic symplectic manifold. Consider a topological twist of M-theory on $\mathbb{R} \times X$ in the presence of a stack of N one-dimensional topological branes supported on $\mathbb{R} \times p$, for some point $p \in X$.

The theorem below is not a global theorem about this M-theory background. It is the formal local model obtained by completing at the brane and separating the local directions into one brane direction, one additional topological direction, and two holomorphic symplectic normal directions:

$$\mathbb{R}_{\mathrm{brane}} \times \mathbb{R}_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}.$$

The extra topological direction is the local topological direction normal to the brane inside the mixed spacetime, while the formal holomorphic disk is the completed normal holomorphic symplectic surface at p.

The local model contains two algebras:

- The algebra $\mathcal{A}_N$ of operators on the stack of N branes supported on $\mathbb{R} \times p$.
- The algebra $\mathcal{B}$ of local operators of the closed mixed Hamiltonian sector on $\mathbb{R} \times X$. Operator product expansions in the direction of $\mathbb{R}$ equip $\mathcal{B}$ with the structure of an associative algebra.

The physical relation between the two algebras is probe-theoretic. The brane algebra records what the closed background does to a finite stack of probes. The closed algebra records infinitesimal Hamiltonian deformations of the formal symplectic neighbourhood. In the large-N limit single traces retain precisely the primitive closed modes; finite-rank trace identities disappear.

The guiding large-N statement, before the sector restrictions and completions are imposed, is:

Conjecture 0.1.2. As $N \to \infty$, the appropriate connected open brane observables are controlled by the Koszul dual of the closed Hamiltonian observable algebra.

Remark 0.1.3. Witten's Chern-Simons model [4] supplies the open-string algebra used for operator-product calculations.

This paper isolates one local mixed holomorphic-topological example and verifies three local statements: the reduced derived Poisson center, the PBW special-fibre trace comparison, and the degree-zero Hamiltonian boundary bracket. The quantum discussion formulates the natural $\hbar$-adic Hamiltonian upgrade, proves the finite-N radial-parts identity for Weyl/Capelli generators, and computes the formal odd-power Moyal target. The graph-realization problem is the analytic BV/Feynman statement that the algebraic boundary bracket and its higher Moyal corrections arise as actual renormalized closed-open graph weights, with propagator, counterterms, anomaly cancellation, and brane-defect kernel fixed. Costello's renormalized BV formalism supplies the general graph/RG package; Costello–Li prove the flat odd-dimensional BCOV/HCS case. The remaining specialization to this Hamiltonian sector is formulated as Problem 0.2.74.

0.1.2 Mixed holomorphic-topological strings

A local string theory in this paper means the part of string field theory visible to local operators. The closed fields form a BV theory; local observables form a factorization algebra; a brane contributes an open cyclic field theory and a bulk-to-boundary map. The words topological and holomorphic describe how these local operators vary in the spacetime directions. This is a local string-field-theoretic use of the word "string", not a definition of a global theory of worldsheet moduli, amplitudes on all backgrounds, or a completed quantum anomaly theory.

Gwilliam–Williams [5] are the precedent for formulating a holomorphic string in the Costello BV/factorization formalism. Their object is the holomorphic bosonic string on a Riemann surface, with Dolbeault fields (γ, β, c, b), a one-loop quantization in the critical complex dimension, and a factorization algebra recovering the BRST cohomology. The object here is not that theory and not a replacement definition. It is a formal local target-space sector: de Rham/Floer directions, a holomorphic symplectic formal disk, Hamiltonian cotangent BF fields, and a brane large-N operator algebra.

The topological word refers to locally constant factorization in the de Rham/Floer directions. The holomorphic word refers to the Dolbeault presentation: the kinetic operator contains $\bar{\partial}$, the fields

are holomorphic jets after local formal completion, and anti-holomorphic translations act by cochain-homotopic zero. A holomorphic theory may acquire further topological descent after quantization; that extra descent is not part of the definition used here.

Definition 0.1.4 *(Working local topological factorization sector).* Let M be a smooth manifold. A local topological closed-string sector on M is a classical BV theory whose local observables form a locally constant factorization algebra on M: for inclusions of sufficiently small disks $U \subset V$, the factorization structure map

$$\mathrm{Obs}^{\mathrm{cl}}(U) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(V)$$

is a quasi-isomorphism. In this paper, attaching branes to such a sector means specifying a cyclic open BV theory and the closed-open operations actually constructed. In a de Rham model the closed fields carry the differential d_M plus the BV linearized differential, and the locally constant factorization product records collision of operators up to isotopy.

Remark 0.1.5. The exact A-model factor used below is a local topological sector in this sense: the self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$ is represented by the de Rham complex $\Omega^*(\mathbb{R})$, and its local observables are insensitive to the metric geometry of the interval. This is the topological part of the mixed holomorphic-topological example.

Definition 0.1.6 *(Working local holomorphic factorization sector).* Let Y be a complex manifold. A local holomorphic sector on Y is a classical BV theory whose fields are Dolbeault complexes on Y, whose differential has the form $\bar{\partial}$ plus the BV linearized differential, whose interactions are holomorphic polydifferential operators, and whose local observables form a holomorphic factorization algebra. Here holomorphic factorization means that, on coordinate polydisks, the factorization products and restriction maps are represented by holomorphic polydifferential operations and every anti-holomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically on observables. In the target-space Kodaira-Spencer examples the fields are Dolbeault-valued polyvectors; in the holomorphic symplectic examples below they are Hamiltonian polyvectors after removing constants. In this paper branes are treated through cyclic open BV theory and the bulk-to-boundary operations constructed below.

Remark 0.1.7. BCOV/Kodaira-Spencer theory is the guiding threefold example [1]: its fields are Dolbeault-valued polyvectors, and the Calabi-Yau volume form supplies the divergence operator. In the holomorphic symplectic surface model used here, the analogous divergence-free vector fields are represented locally by Hamiltonians modulo constants. The surface model is not a truncation of threefold BCOV theory.

Remark 0.1.8. The classical local observable theory underlying Gwilliam–Williams is an example of a pure holomorphic BV/factorization sector on a Riemann surface, with additional worldsheet ghost, BRST, and critical-dimension quantum structure. Their critical-dimension and BRST-cohomology theorems are not imported here. The model below is a different formal local target-space sector whose holomorphic sector is Hamiltonian BF, or the holomorphic symplectic Kodaira-Spencer analogue, and whose topological sector is de Rham/Floer. In particular, the critical complex dimension in Gwilliam–Williams is a worldsheet ghost-anomaly statement for the holomorphic bosonic string; it is not a constraint on the local $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian sector studied here.

Definition 0.1.9 *(Working classical local mixed holomorphic-topological sector).* Let M be a smooth manifold and let Y be a complex manifold. A classical local mixed holomorphic-topological closed sector on $M \times Y$ consists of:

(i) a sheaf of cochain complexes of fields $\mathcal{E}$ which, on product coordinate opens $D \times B$, has the form

$$\mathcal{E}(D \times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B, \mathcal{V})$$

for a graded holomorphic vector bundle or formal holomorphic vector space $\mathcal{V}$;

(ii) a degree -1 local BV pairing on $\mathcal{E}$ and a differential

$$Q = d_M + \bar{\partial}_Y + Q_0,$$

where Q_0 is holomorphic over Y and is linear over de Rham forms on M;

(iii) a degree-zero local functional $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E})$, built from differential operators in the M directions and holomorphic polydifferential operators in the Y directions, satisfying the classical master equation

$$QI + \frac{1}{2}\{I, I\} = 0;$$

(iv) the associated classical observables

$$\mathrm{Obs}^{\mathrm{cl}}(U) = \left(\widehat{\mathrm{Sym}}(\mathcal{E}^!_c(U)), Q_{\mathrm{obs}}\right), \qquad Q_{\mathrm{obs}} = Q^! + \{I, -\},$$

with the standard factorization product for disjoint supports. Here $\mathcal{E}^!_c(U)$ denotes compactly supported continuous linear observables, i.e. the density-dual cosheaf of fields; extension by zero gives the covariant structure maps used by the factorization algebra.

These data must satisfy the mixed locality condition. For every inclusion of small disks $D \subset D'$ in M and every holomorphic polydisk $B \subset Y$, the structure map

$$\mathrm{Obs}^{\mathrm{cl}}(D \times B) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(D' \times B)$$

is a quasi-isomorphism. In holomorphic coordinates on B, each anti-holomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically on observables: there is a degree -1 derivation η_i with

$$[Q_{\mathrm{obs}}, \eta_i] = \mathcal{L}_{\partial/\partial\bar{z}_i}.$$

A mixed brane for such a sector is supported on $N \times Z$, where $N \subset M$ is a smooth submanifold and $Z \subset Y$ is a holomorphic submanifold or a formal holomorphic neighbourhood of one, and is specified by an object $\mathcal{F}$ of the local holomorphic brane category on Z. Its open fields have the derived form

$$\Omega^*(N)\widehat{\otimes}\mathrm{RHom}_Y(\mathcal{F}, \mathcal{F}) \otimes \mathrm{End}(V)[1],$$

with a cyclic trace. For a smooth brane this recovers the Dolbeault model $\Omega^*(N)\widehat{\otimes}\Omega^{0,*}(Z, \mathcal{W})$; for a skyscraper brane it produces the Ext algebra. A full closed-open mixed datum would further require a morphism of factorization algebras

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed}}\big|_{N\times Z} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}),$$

from closed observables to the Poisson/BV factorization center of the open brane observables, compatible with the BV brackets and factorization products. In the Hamiltonian model below we construct the degree-zero Hamiltonian current shadow of this map; the full closed CE/cotangent factorization-center lift is Problem 0.2.25.

Remark 0.1.10. A quantum mixed holomorphic-topological string datum would be the corresponding $\hbar$-adic factorization algebra obtained from a renormalized BV action satisfying the quantum master equation. This paper constructs no such quantum datum for the Hamiltonian model. It proves the classical local closed-open Hamiltonian theorem and formulates the quantum upgrade in Problem 0.2.64.

Remark 0.1.11 *(Locality and unitarity).* Locality in this paper is operational locality of observables, not Lorentzian finite propagation. If insertions have disjoint supports, the factorization product records that the measurements can be performed independently. In the topological directions translations are Q-exact, so the position of an insertion is invisible until a collision occurs. In the holomorphic normal directions the theory is formal at the brane point; principal parts are transverse delta-currents, or derivatives of them, supported at the defect.

The model is not unitary in the ordinary Lorentzian sense. It is complex, BV, ghost-valued, and topological-holomorphic; it has no positive Hilbert space inner product or reflection-positivity statement. The consistency conditions used here are the classical master equation, and in the quantum problem the quantum master equation together with factorization/OPE associativity. If this local sector comes from a unitary untwisted parent, that unitarity belongs to the parent theory, not to the protected BV sector by itself.

Definition 0.1.9 separates three levels. A pure topological sector is locally constant in all real directions; in examples this includes the A-model local/Floer direction and the Chern-Simons-type open field theory on a topological brane. A pure holomorphic sector is Dolbeault in its complex directions and produces a holomorphic factorization algebra; Gwilliam–Williams' holomorphic bosonic string is a worldsheet example, while BCOV/Kodaira-Spencer gives the guiding target-space closed-string example. The Hamiltonian BF complex below is the formal local target-space sector used in this paper. A mixed holomorphic-topological sector lives on $M \times Y$ with locally constant factorization in M and holomorphic factorization in Y. The A/B terminology used below is one source of such examples: the A-side supplies the de Rham/Floer direction and the B-side supplies the Dolbeault or formal holomorphic direction.

0.1.3 The local mixed model

The paper studies the product local model

$$\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}.$$

The theorem is not a global statement about all twisted M-theory backgrounds. It is the formal local calculation obtained after completing at a brane $\mathbb{R} \times p$ and taking the holomorphic symplectic normal disk at $p \in \mathbb{C}^2$, with

$$\omega = dz_1 \wedge dz_2.$$

The closed sector is the Hamiltonian Kodaira-Spencer/BF theory on this mixed space, not a pure A-model or pure B-model. The open sector is the cyclic open string field theory on the mixed brane: topological along $\mathbb{R}$ and formal holomorphic in the normal $\mathbb{C}^2$ directions.

Remark 0.1.12 *(Global obstruction).* The Hamiltonian identification used here is local. On a holomorphic symplectic surface X, a holomorphic vector field v is symplectic when

$$\alpha_v = -\iota_v \omega$$

is a closed holomorphic one-form, and it is Hamiltonian only after choosing f with $df = \alpha_v$, in the convention $\iota_{X_f}\omega = -df$. Thus globalizing the Hamiltonian sector requires either a sheaf of local Hamiltonians or vanishing of the corresponding holomorphic de Rham obstruction. The formal disk has no such obstruction by the holomorphic Poincare lemma; a general X need not.

PROPOSITION 0.1.13 *(Global local-Hamiltonian restriction).* Let (X, ω) be a connected holomorphic symplectic surface and let $p \in X$. Put

$$\mathcal{H}_X = \mathcal{O}_X/\mathbb{C}_X, \qquad \widehat{\mathfrak{h}}_{X,p} = \widehat{\mathcal{O}}_{X,p}/\mathbb{C}.$$

The differential identifies $\mathcal{H}_X$ with the sheaf of closed holomorphic one-forms, hence with the sheaf of holomorphic symplectic vector fields by $\alpha = -\iota_v\omega$. For every $p \in X$ there is a Lie algebra restriction

$$H^0(X, \mathcal{H}_X) \longrightarrow \widehat{\mathfrak{h}}_{X,p}.$$

The formal neighbourhood $(\widehat{X}_p, \omega)$ defines the same stable Hamiltonian brane sector as the local model of Theorem 0.1.17, after choosing a formal Darboux coordinate system. The resulting boundary map

$$\widehat{\mathfrak{h}}_{X,p} \longrightarrow H^0_{\mathrm{Ham}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{brane}})$$

is independent of this choice up to the natural action of the formal symplectomorphism group. The subspace coming from global Hamiltonian functions is the image of

$$H^0(X, \mathcal{O}_X)/\mathbb{C} \longrightarrow H^0(X, \mathcal{H}_X).$$

The obstruction for a locally Hamiltonian symplectic vector field to come from a global Hamiltonian function is its period class in $H^1(X, \mathbb{C}_X)$.

Proof. The holomorphic Poincare lemma gives an isomorphism

$$d : \mathcal{O}_X/\mathbb{C}_X \xrightarrow{\sim} \Omega^1_{X,\mathrm{cl}}.$$

The holomorphic symplectic form identifies a symplectic vector field v with the closed one-form $\alpha_v = -\iota_v\omega$. Thus a section of $\mathcal{H}_X$ is exactly a local Hamiltonian primitive, modulo local constants, and the Poisson bracket is independent of the local constants. Taking the germ at p and completing gives the displayed restriction to $\widehat{\mathfrak{h}}_{X,p}$.

The formal holomorphic Darboux theorem identifies the completed local symplectic algebra at p with $\mathbb{C}[[z_1, z_2]]$ equipped with $dz_1 \wedge dz_2$. Under this identification Hamiltonians modulo constants become $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$, and the point-brane Ext algebra is the coordinate-free exterior algebra $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda^* T_p X$, with cyclic trace fixed by ω_p. Hence the open BV reduction and the Hamiltonian boundary evaluation are exactly the formal local construction after choosing Darboux coordinates. A different Darboux coordinate system differs by a formal symplectomorphism, which acts on Hamiltonians and on the normal coordinate algebra; the substitution map $z_i \mapsto \phi_i$ is equivariant for this action.

The exact sequence

$$0 \to \mathbb{C}_X \to \mathcal{O}_X \to \mathcal{H}_X \to 0$$

gives

$$0 \to H^0(X, \mathcal{O}_X)/\mathbb{C} \to H^0(X, \mathcal{H}_X) \xrightarrow{\partial} H^1(X, \mathbb{C}_X) \to H^1(X, \mathcal{O}_X).$$

For a locally Hamiltonian vector field, δ is the period class of the closed one-form $-\iota_v\omega$. Vanishing of this class is exactly the existence of a single global holomorphic Hamiltonian primitive, unique modulo constants. □

Remark 0.1.14. This is the maximal global statement used in the paper. If $H^1(X,\mathbb{C}) = 0$, global symplectic vector fields which are locally Hamiltonian have global Hamiltonian primitives. If X is Stein, every finite jet needed for a finite local calculation can be lifted to a global holomorphic Hamiltonian, noncanonically: apply Cartan's theorem B to

$$0 \to \mathfrak{m}_p^{k+1} \to \mathcal{O}_X \to \mathcal{O}_X/\mathfrak{m}_p^{k+1} \to 0.$$

This does not imply surjectivity onto $\widehat{\mathcal{O}}_{X,p}$. If X is compact connected, then $H^0(X,\mathcal{O}_X)/\mathbb{C} = 0$; abelian surfaces still have nonzero locally Hamiltonian translations, obstructed by their period classes. A global cochain-level Poisson factorization-center map or a global quantum deformation-quantization statement requires additional descent, anomaly, and background-choice data.

The closed Hamiltonian coefficient Lie algebra is the reduced Tate vector space

$$\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1.$$

We write a class by its unique zero-constant Taylor representative. The bracket is the projected Poisson bracket

$$\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g \mod \mathbb{C}\cdot 1.$$

Thus $\{z_1, z_2\} = 0$ in $\widehat{\mathfrak{h}}$, although the unreduced Poisson bracket of the chosen representatives is 1. The cotangent BF field contributes the continuous dual, so the residual local closed Lie algebra is

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1].$$

On product opens in $\mathbb{R}^2 \times \mathbb{C}^2$ the closed fields have the mixed form

$$\Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

with differential $d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}$ and BF interaction $\frac{1}{2}\int \langle \beta, \{\alpha,\alpha\}\rangle dz_1 dz_2$. Proposition 0.2.1 checks that this is a local mixed holomorphic-topological BV sector in the sense above.

The open brane is the product of the exact local A-brane $\mathbb{R} \subset T^*\mathbb{R}$ and the B-brane $\mathcal{O}_p$ at the formal holomorphic point. For a cyclic Calabi-Yau A_∞ category, the open field of an object $\mathcal{F}$ is $\mathrm{RHom}(\mathcal{F},\mathcal{F})[1]$ and the formal Chern-Simons action is

$$S(a) = \sum_{n\geq 1} \frac{1}{n+1}\mathrm{Tr}\big(a\, m_n(a,\dots,a)\big), \qquad a \in \mathrm{RHom}(\mathcal{F},\mathcal{F})[1].$$

This is the algebraic core of Witten's open string field theory [4] and of the AKSZ packaging [6]. In the present local model

$$\mathrm{RHom}_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p,\mathcal{O}_p) \cong \Lambda(\varepsilon_1,\varepsilon_2),$$

and the ghost-zero open field is

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2.$$

The ghost-zero action is the first-order gauged matrix model

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1,\phi_2]\big),$$

where ϕ_1, ϕ_2 are adjoint scalar normal coordinates and A imposes the moment-map equation $[\phi_1,\phi_2] = 0$.

0.1.4 Hamiltonian sector and stable large N

Everything below is over $\mathbb{C}$. The phrase *formal local model* means the completed Taylor model at the chosen point of $\mathbb{C}^2$; equivalently, one may work polynomially degree by degree and then complete at the augmentation ideal. The symbol $\widehat{\mathbf{S}}$ denotes this completed symmetric algebra, and all Chevalley-Eilenberg cochains are continuous cochains for the same filtration.

Shifts follow the BV convention used in the field descriptions: an element of $V[1]$ has parity reversed and cohomological degree one lower than the corresponding element of V. The notation $\mathbf{S}(V)$ always means the free graded-commutative algebra on the graded vector space V.

Let

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1, \qquad \mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}\cdot 1, \qquad \mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}.$$

Constants are removed because a constant Hamiltonian has zero Hamiltonian vector field. We use zero-constant Taylor representatives for calculations, but the Lie bracket is always the Poisson bracket followed by projection modulo constants. On the brane side $\mathrm{Tr}_N(1) = N$ is the empty trace, a central scalar depending on the rank. We do not quotient the associative operator algebra by its unit. The Hamiltonian comparison uses the connected stable trace sector: the unit is retained in the symmetric algebra, but the empty trace is not a primitive generator and constants project to zero in the Hamiltonian Lie/vector quotient. If T is a positive word-degree trace expression, $\overline{T}$ denotes its connected trace class; constants have zero connected class.

Remark 0.1.15 *(Scalar Hamiltonian and center-of-mass term).* The quotient by constants is load-bearing. In the reduced Hamiltonian algebra of the formal disk,

$$\{z_1, z_2\} = 1 \equiv 0 \quad \text{in } \bar{A}.$$

On the unreduced brane phase space, however,

$$\{\mathrm{Tr}(\phi_1), \mathrm{Tr}(\phi_2)\} = N.$$

This is the center-of-mass symplectic pair of the stack. Every bracket statement below involving linear Hamiltonians is therefore a statement after projection to the Hamiltonian quotient by scalar trace classes. The same term reappears quantum mechanically as the Capelli shift in Remark 0.2.61.

At a fixed point of the brane line the reduced BV algebra is generated by two even matrices ϕ_1, ϕ_2 and the odd antifield $\psi = A^*$, with

$$Q\phi_1 = Q\phi_2 = 0, \qquad Q\psi = [\phi_1, \phi_2].$$

Thus the moment-map equation is imposed homologically. Boundary terms on $\mathbb{R}$ are discarded by taking compactly supported variations, or equivalently by working on an interval with fixed boundary values and passing to local operators in the interior.

The term *stable large-N* means the degreewise stable value of the invariant algebras under block restriction as $N \to \infty$. Inside a fixed finite matrix algebra all finite-N trace identities remain present. In the stable limit, Procesi–Razmyslov invariant theory [18, 19] says that invariant functions of two matrices are freely generated by cyclic trace words. The differential $Q\psi = [\phi_1, \phi_2]$ then makes adjacent swaps exact, so ghost-zero cyclic words reduce to ordinary commutative monomials

$$\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \qquad k + l > 0.$$

The primitive one-ψ Koszul homology supplies a second nonconstant family $\Psi_{k,l}$, obtained by inserting one ψ and symmetrising over all words with k copies of ϕ_1 and l copies of ϕ_2. The scalar class $\mathrm{Tr}(\psi)$ is excluded from the Hamiltonian comparison because constants have been removed from $\bar{A}$; higher ψ-count homology lies outside this Hamiltonian sector.

The large-N principle is the string-field-theoretic form of the Loday-Quillen-Tsygan principle [2, 3]. Open strings on a stack produce matrix algebras; closed modes appear in the primitive stable part. The theorem below verifies this principle directly in the present connected Hamiltonian trace sector.

Theorem 0.1.16 *(Loday-Quillen-Tsygan).* Let A be a unital associative algebra over a field of characteristic zero. The stable Chevalley-Eilenberg homology of $\mathfrak{gl}_\infty(A)$ is the free graded-commutative algebra generated by shifted cyclic homology:

$$H_*^{\mathrm{CE}}\big(\mathfrak{gl}_\infty(A)\big) \cong \mathbf{S}\big(HC_{*-1}(A)\big).$$

Here this theorem is a template rather than a direct input: stable matrix constructions are controlled by cyclic words, while the particular reduced model is computed by Procesi–Razmyslov invariant theory and the explicit Koszul differential above. No global HC_{*-1} convention is imported into the main theorem; cyclic homology supplies the guiding stable-trace analogy, not the proof of the Hamiltonian sector.

0.1.5 *The concrete dictionary*

We write $\bar{A}_{\mathrm{desc}}$ for an abstract copy of $\bar{A}$ whose basis vector labelled by $z_1^k z_2^l$, with $k + l > 0$, is represented on the brane by the first descendant $\Psi_{k,l}$. On the closed side the cotangent field gives the continuous Taylor dual $\mathfrak{h}^\vee$. Choosing the monomial Taylor basis identifies $\mathfrak{h}^\vee$ with $\bar{A}_{\mathrm{desc}}$ as a filtered vector space. The closed module structure remains the coadjoint module structure.

For $k, l \geq 0$ and $k + l > 0$, set

$$H_{k,l} = z_1^k z_2^l \in \bar{A}, \qquad E_{k,l} = \delta_{k,l}[1] \in \mathfrak{h}^\vee[1],$$

where $\delta_{k,l}$ is the coefficient functional dual to the Taylor monomial $z_1^k z_2^l$:

$$\delta_{k,l}(z_1^a z_2^b) = \delta_{k,a}\delta_{l,b}.$$

On the brane side set

$$O_{k,l} = \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)},$$

and let $\Psi_{k,l}$ be the symmetrised trace with one ψ, k copies of ϕ_1, and l copies of ϕ_2. The theorem proves that

$$H_{k,l} \longmapsto O_{k,l}, \qquad E_{k,l} \longmapsto \Psi_{k,l}$$

extends to an isomorphism of the connected stable Hamiltonian descendant sector with

$$\mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]) \xrightarrow{\sim} \mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H}$$

in the sector considered here. On the closed side the symbols $H_{k,l}$, $E_{k,l}$ form the degreewise polynomial PBW generators inside the special fibre of $U_\hbar(\mathfrak{g})$.

Thus every later step has a concrete target: construct the symbols $O_{k,l}$, $\Psi_{k,l}$ from the open fields, construct $H_{k,l}$, $E_{k,l}$ from Hamiltonians and cotangent Hamiltonians, prove that commutators of words reduce to commutative monomials, and check that the bracket coefficient is $(kn - lm)$.

0.1.6 *The local theorem*

The theorem has three layers. First, the open calculation says that the ghost-zero brane theory is the algebra of invariant functions on commuting matrix pairs, together with the first Koszul descendants of the commuting-pair equation. Second, the closed calculation says that the same generators are the PBW symbols of the Hamiltonian cotangent Lie algebra. Third, the boundary map is not an abstract isomorphism: it is the physical operation of evaluating a Hamiltonian on the matrix-valued normal coordinates of the brane.

THEOREM 0.1.17 *(PBW special fibre for the stable Hamiltonian trace sector).* In the formal Hamiltonian sector of the local mixed holomorphic-topological BV string-field model on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $\mathbb{R} \times p$ constructed in Proposition 0.2.1, the stable large-N connected open Hamiltonian trace sector freely generated by the nonempty primitive degree-zero Hamiltonian traces and the nonconstant primitive one-antifield Koszul classes is

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H} \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]\big).$$

The classical local operators in the formal Hamiltonian cotangent BF sector are the continuous Chevalley-Eilenberg algebra

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}} \cong C^{\bullet}_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^{\vee}_{\mathrm{cont}}[-1]\big).$$

Its Koszul dual is the completed graded enveloping algebra

$$\big(\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}}\big)^{!} \simeq \widehat{U(\mathfrak{g})}.$$

The PBW/order Rees algebra $U_\hbar(\mathfrak{g})$ has completed special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^{\vee}[1]\big).$$

Its degreewise polynomial subalgebra $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^{\vee}[1])$ identifies with the open stable Hamiltonian-sector algebra after choosing the monomial-dual basis $\mathfrak{h}^{\vee} \cong \bar{A}_{\mathrm{desc}}$. This is a filtered graded-commutative PBW special-fibre product identification, not a deformation-level identification of the descendant module actions. The open-side deformation is checked below on degree-zero Hamiltonian generators. On those generators the identification is induced by the connected bulk-to-boundary evaluation map

$$\partial_{\mathrm{bb}}(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)},$$

which sends a Hamiltonian Lie/PBW mode, not a closed CE cochain representative, to the corresponding brane Hamiltonian and intertwines Poisson brackets. The shifted PBW special-fibre symbols are labelled by the nonconstant first Koszul descendants $\Psi_{k,l}$; this is not an $\mathfrak{h}$-module identification. On monomials,

$$z_1^k z_2^l \longleftrightarrow \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \qquad k + l > 0,$$

the first bracket is

$$\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})},$$

matching the Poisson bracket on $\bar{A}$. If a displayed exponent is negative, the coefficient is already zero; if the displayed monomial is constant, its connected trace class is zero.

The proof uses the formal local model alone. A complete closed-open mixed datum would refine the bracket-compatible Hamiltonian boundary evaluation to a Poisson factorization-center morphism

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed}}\big|_{\mathbb{R}\times p} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}).$$

Proposition 0.2.19 constructs the degree-zero Hamiltonian current shadow of this map. The full closed CE/cotangent lift is Problem 0.2.25. The quantum calculation later fixes the formal Weyl/Moyal target; the analytic problem is to identify Costello's renormalized BV graph package with this Hamiltonian BF sector and to recover the first and third graph normalizations.

0.2 The Local Model

0.2.1 Setup

Let

$$M = \mathbb{R}^2, \qquad Y = \mathbb{C}^2, \qquad X = M \times Y.$$

The real factor M is the topological direction and the complex factor Y is the holomorphic direction; all Hamiltonian calculations are completed at the formal neighbourhood of $p = 0 \in Y$. We use the exact A-brane given by the standard line $\mathbb{R} \subset M$ with trivial brane data and the B-brane $\mathcal{O}_p \in \mathrm{Perf}(Y)$. Consider the mixed brane

$$\mathbb{R} \times p \rightarrowtail \mathbb{R}^2 \times \mathbb{C}^2$$

for some point $p \in Y$. We take $p = 0$ in coordinates. The holomorphic factor carries the standard holomorphic symplectic form

$$\omega = dz_1 \wedge dz_2.$$

Thus the model is topological along $\mathbb{R}^2$ and holomorphic Hamiltonian along $\mathbb{C}^2$.

Proposition 0.2.1 *(Local mixed model).* The formal local Hamiltonian data used here define a closed mixed holomorphic-topological BV sector and a compatible cyclic open brane BV theory. The constructed closed-open operation is the degree-zero Hamiltonian boundary evaluation. The full Poisson factorization-center morphism is Problem 0.2.25, not part of this proposition.

Proof. On product opens $D \times B$ in the completed formal neighbourhood of $\mathbb{R}^2 \times \{p\} \subset \mathbb{R}^2 \times \mathbb{C}^2$, the closed fields are

$$\mathcal{E}_{\mathrm{cl}}(D \times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big).$$

Degreewise polynomial calculations use the dense subspace $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The differential is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, so $Q_0 = 0$ in Definition 0.1.9. The BV pairing is the local pairing on compactly supported fields, obtained by multiplying the form components, integrating over $D \times B$ against $dz_1\, dz_2$, and using the continuous evaluation pairing $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}} \otimes \widehat{\mathfrak{h}} \to \mathbb{C}$. If $\alpha \in \Omega^p\widehat{\otimes}\widehat{\mathfrak{h}}[1]$ and $\beta \in \Omega^q\widehat{\otimes}\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]$, then $|\alpha| = p - 1$ and $|\beta| = q - 2$; the top-form condition $p + q = 4$ makes the pairing degree -1.

The free quadratic term

$$S_{\mathrm{free}} = \int_{D\times B} \big\langle \beta, D\alpha \big\rangle dz_1\, dz_2$$

is the BV Hamiltonian for the differential D. The interaction in Definition 0.1.9 is only

$$I_{\mathrm{BF}} = \frac{1}{2}\int_{D\times B} \big\langle \beta, \{\alpha, \alpha\} \big\rangle dz_1\, dz_2.$$

The total BF action is $S_{\text{free}} + I_{\text{BF}}$. The Poisson bracket on $\bar{A}$ is the holomorphic bracket determined by $dz_1 \wedge dz_2$. Since D is a derivation of this bracket and the bracket satisfies Jacobi,

$$DI_{\text{BF}} + \frac{1}{2}\{I_{\text{BF}}, I_{\text{BF}}\} = 0,$$

so the total BF action satisfies the classical master equation without double-counting the free differential.

Mixed locality is the Cartan homotopy calculation on the interacting observables. For a constant vector field v in the $\mathbb{R}^2$ direction, $[D, \iota_v] = \mathcal{L}_v$; for an anti-holomorphic coordinate vector field $\partial/\partial\bar{z}_i$, $[D, \iota_{\partial/\partial\bar{z}_i}] = \mathcal{L}_{\partial/\partial\bar{z}_i}$. The interaction is built from the de Rham wedge product and holomorphic polydifferential operators, so the contractions are derivations of the local observable algebra and commute with $\{I_{\text{BF}}, -\}$ up to the displayed Lie derivative. Thus $[Q_{\text{obs}}, \iota_v] = \mathcal{L}_v$ and $[Q_{\text{obs}}, \iota_{\partial/\partial\bar{z}_i}] = \mathcal{L}_{\partial/\partial\bar{z}_i}$ on local observables. The de Rham Poincare lemma in the D directions gives the quasi-isomorphism for small disk inclusions, and the Dolbeault Poincare lemma gives the holomorphic dependence in the B direction.

The brane is $N \times Z = \mathbb{R} \times \{p\}$, with Z understood as the formal holomorphic neighbourhood at p. Its open fields are

$$\Omega^*(\mathbb{R}) \otimes \mathbf{Ext}^*_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \otimes \mathfrak{gl}_N[1],$$

and Proposition 0.2.2 identifies the Ext algebra with $\Lambda(\varepsilon_1, \varepsilon_2)$. This is the derived $\mathrm{RHom}_Y(\mathcal{O}_p, \mathcal{O}_p)$ allowed in Definition 0.1.9, not an ordinary Dolbeault algebra on the point. The trace on $\mathfrak{gl}_N$ and the Berezin trace on this exterior algebra give the cyclic open string field theory. Construction 0.2.32 and Theorem 0.2.58 construct the degree-zero Hamiltonian boundary component and prove its bracket compatibility. The full Poisson factorization-center closed-open map is Problem 0.2.25, not a theorem proved here. □

0.2.2 *Open mixed brane states*

The open string states are the tensor product of the topological brane direction and the formal holomorphic normal direction.

- *Topological factor.* In the exact self-Floer model for the graded, spin zero-section $\mathbb{R} \subset T^*\mathbb{R} \cong \mathbb{R}^2$, we use the local exact Fukaya model near a compact interval in the brane. A small compactly supported Hamiltonian perturbation identifies $CF^*(\mathbb{R}, \mathbb{R})$ with the Morse complex on that interval, and passage to local operators in the interior is represented by the de Rham complex $\Omega^*(\mathbb{R})$ with compactly supported variations. No wrapped generators at infinity are included in this local model.
- *Holomorphic factor.* Put affine coordinates at $p = 0$ and write $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $\mathcal{O}_p = R/\mathfrak{m}$. The skyscraper sheaf has open string states

  $$\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m}),$$

  computed in Proposition 0.2.2.

PROPOSITION 0.2.2 *(Affine skyscraper self-Ext).* Let $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $k = R/\mathfrak{m}$. Then the Yoneda algebra of the skyscraper sheaf at the origin is

$$\mathbf{Ext}^*_R(k, k) \cong \Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2),$$

where $|\varepsilon_i| = 1$. In particular this is the free graded-commutative algebra on two odd generators.

Proof. The pair (z_1, z_2) is a regular sequence in the polynomial ring R, so the affine Koszul complex

$$K = \left(0 \to R\, e_{12} \xrightarrow{d_2} R\, e_1 \oplus R\, e_2 \xrightarrow{d_1} R \to 0\right)$$

is a free resolution of k, with

$$d_1(ae_1 + be_2) = az_1 + bz_2, \qquad d_2(e_{12}) = z_2 e_1 - z_1 e_2.$$

Applying $\mathrm{Hom}_R(-, k)$ gives

$$0 \to k \xrightarrow{0} k\,\varepsilon_1 \oplus k\,\varepsilon_2 \xrightarrow{0} k\,\varepsilon_{12} \to 0,$$

because z_1 and z_2 act by zero on k. Hence the graded vector space of Ext groups is k in degree 0, $k\varepsilon_1 \oplus k\varepsilon_2$ in degree 1, and $k\varepsilon_{12}$ in degree 2.

The product is the Yoneda product. The Koszul resolution is the exterior dg-algebra $R \otimes \Lambda(e_1, e_2)$ with $d(e_i) = z_i$; its diagonal sends e_i to $e_i \otimes 1 + 1 \otimes e_i$. Dualising this diagonal gives $\varepsilon_i^2 = 0$ and $\varepsilon_1\varepsilon_2 = -\varepsilon_2\varepsilon_1$, while $\varepsilon_1\varepsilon_2$ spans the top Ext group. Thus the Yoneda algebra is $\Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2)$. □

0.2.3 *Open mixed brane field theory*

The open theory is the minimal local field theory which remembers the brane's two normal holomorphic directions. The two Ext generators are odd because they are derived normal directions of a point brane; after the BV shift their ghost-zero coefficients become the two ordinary matrix coordinates ϕ_1, ϕ_2. The one-form coefficient is the gauge field A along the brane. Thus the formal Chern-Simons action is the mechanism which turns the Ext algebra into first-order matrix mechanics.

The total space of fields (in the Batalin-Vilkovisky formalism) is

$$\alpha \in \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1]$$

with action

$$S(\alpha) = \int_{\mathbb{R}\times\mathbb{C}^{0|2}} \frac{1}{2}\,\mathrm{Tr}(\alpha\, d\alpha) + \frac{1}{3}\,\mathrm{Tr}\,\alpha^3.$$

Here $d = d_{\mathrm{dR}}$; the Ext algebra $\Lambda(\varepsilon_1, \varepsilon_2)$ has zero differential. The integral over $\mathbb{C}^{0|2}$ is the Berezin integral $\int d\varepsilon_1\, d\varepsilon_2$, which picks out the top component in the odd directions.

The cyclic trace is

$$\mathrm{Tr}_{\mathrm{open}}(\eta \otimes u \otimes M) = \int_{\mathbb{R}} \eta\, [\varepsilon_1\varepsilon_2]u\ \mathrm{Tr}_N(M),$$

where $[\varepsilon_1\varepsilon_2]u$ denotes the coefficient of the top Ext class. The top Ext class is the residue orientation dual to $dz_1 \wedge dz_2$. The de Rham integral has degree -1 and the Ext residue has degree -2; after the field shift by $[1]$, the induced BV pairing has degree -1.

Construction 0.2.3 *(The open field from the Ext basis).* Write $\varepsilon_{12} = \varepsilon_1\varepsilon_2$. Before imposing ghost number, an open field is a finite sum

$$\alpha = a_0 + a_1\varepsilon_1 + a_2\varepsilon_2 + a_{12}\varepsilon_{12},$$

where each a_I is a de Rham form on $\mathbb{R}$ with values in $\mathfrak{gl}_N$, shifted by $[1]$. The ghost-number-zero components are exactly

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N, \qquad \phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and they appear as

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2.$$

The Berezin integral kills every monomial except the coefficient of ε_{12}, so every term in the ghost-zero action is obtained by multiplying the displayed three summands and keeping exactly one real one-form and exactly the product $\varepsilon_1\varepsilon_2$.

Remark 0.2.4. The shift [1] acts on the total cohomological degree: a form of de Rham degree p and Grassmann degree q has total degree $p + q$, which after the shift sits in ghost number $p + q - 1$. In particular, the 1-form component of α (de Rham degree 1, Grassmann degree 0) sits in ghost number 0 and is the bosonic gauge field. The scalar components proportional to ε_i (de Rham degree 0, Grassmann degree 1) also sit in ghost number 0 and represent the motion of the brane in the two normal directions.

0.2.3.1 GHOST-ZERO OPEN FIELD CONTENT

The ghost-number-zero part of the BV field is

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2,$$

where $A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N$ and $\phi_i \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N$.

LEMMA 0.2.5. With the Berezin convention $\int_{\mathbb{C}^{0|2}} \varepsilon_1\varepsilon_2 = 1$, the restriction of the BV action to α_0 is, up to a boundary term on $\mathbb{R}$,

$$S(\alpha_0) = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Proof. Only monomials containing one real one-form and the factor $\varepsilon_1\varepsilon_2$ survive the integration over $\mathbb{R} \times \mathbb{C}^{0|2}$. In the quadratic term this leaves

$$\frac{1}{2}\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 - \phi_2\, d\phi_1\big),$$

where the sign is the Koszul sign from interchanging the two odd generators. Since $d\,\mathrm{Tr}(\phi_1\phi_2) = \mathrm{Tr}(d\phi_1\,\phi_2 + \phi_1\, d\phi_2)$, the second summand differs from $\mathrm{Tr}(\phi_1\, d\phi_2)$ by an exact one-form. In the cubic term the six orderings of A, $\phi_1\varepsilon_1$, $\phi_2\varepsilon_2$ reduce by cyclicity of the trace and the odd relation $\varepsilon_2\varepsilon_1 = -\varepsilon_1\varepsilon_2$ to

$$3\,\mathrm{Tr}(A\phi_1\phi_2 - A\phi_2\phi_1).$$

The coefficient $\frac{1}{3}$ in the Chern-Simons action therefore gives exactly $\mathrm{Tr}(A[\phi_1, \phi_2])$. □

LEMMA 0.2.6 *(Dirac reduction of the probe phase space).* Let

$$\mathcal{P}_N = \mathfrak{gl}_N \oplus \mathfrak{gl}_N$$

with coordinates (ϕ_1, ϕ_2) and symplectic form

$$\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2).$$

With the convention $\iota_{X_f}\Omega_N = -df$, simultaneous conjugation by $\mathfrak{gl}_N$ is Hamiltonian with moment map components

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

They satisfy the first-class identity

$$\{\mu_\epsilon, \mu_\eta\} = \mu_{[\epsilon,\eta]}.$$

The action of Lemma 0.2.5 is the corresponding Dirac first-order action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A,$$

and varying A imposes the constrained phase space

$$\mu^{-1}(0)/GL_N = \{[\phi_1, \phi_2] = 0\}/GL_N.$$

Proof. The symplectic form gives

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j.$$

For $v_\epsilon(\phi_i) = [\epsilon, \phi_i]$, cyclicity of the trace gives

$$d\mu_\epsilon(\delta\phi_1, \delta\phi_2) = -\mathrm{Tr}\big(\epsilon([\delta\phi_1, \phi_2] + [\phi_1, \delta\phi_2])\big) = -\Omega_N(v_\epsilon, (\delta\phi_1, \delta\phi_2)).$$

Thus $X_{\mu_\epsilon} = v_\epsilon$. The moment-map bracket is therefore the infinitesimal equivariance identity $\{\mu_\epsilon, \mu_\eta\} = \mu_{[\epsilon,\eta]}$. Substituting $\epsilon = A(t)$ gives

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big),$$

and the A-equation is $[\phi_1, \phi_2] = 0$. □

Proposition 0.2.7 *(Classical BV truncation).* The subspace of ghost-number-zero fields

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$$

with all nonzero ghost-number components set to zero is a consistent classical truncation of the open BV theory. The induced classical action is the first-order gauged matrix model of Lemma 0.2.5.

Proof. Decompose the BV field as $\alpha = \sum_r \alpha^{(r)}$ by ghost number. The differential $d_{\mathbb{R}}$ and the product on $\Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N$ preserve total degree, and the BV action has ghost number zero. Hence every monomial in the Euler-Lagrange expression for a component of ghost number $r \neq 0$ contains at least one field component of nonzero ghost number when evaluated at $\alpha = \alpha_0$. Setting all such components to zero therefore makes the discarded equations vanish. The remaining equation is exactly the variation of $S(\alpha_0)$, computed in Lemma 0.2.5.

Concretely, the full Euler-Lagrange expression is the curvature

$$F(\alpha) = d\alpha + \alpha^2.$$

For $\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$ on a line,

$$F(\alpha_0) = (d\phi_1 + [A, \phi_1])\varepsilon_1 + (d\phi_2 + [A, \phi_2])\varepsilon_2 + [\phi_1, \phi_2]\varepsilon_1\varepsilon_2.$$

The BV pairing keeps exactly real-form degree one and $\varepsilon_1\varepsilon_2$. Hence the displayed three components pair exactly with the variations of A, ϕ_2, ϕ_1 respectively, while variations of nonzero ghost number see complementary curvature components that are zero for α_0 by real-form degree or ε-degree. This is a physical ghost-zero truncation; the residual ordinary GL_N gauge symmetry remains. □

The ghost-number-zero truncation of the BV master action is closed under the classical equations and gives a quantum-mechanical system with a connection form

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

two adjoint scalars

$$\phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and action

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

This action is invariant under

$$\delta_\varepsilon \phi_i = [\varepsilon, \phi_i], \qquad \delta_\varepsilon A = d\varepsilon + [A, \varepsilon],$$

for $\varepsilon \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N$. Indeed the variation of $\mathrm{Tr}(\phi_1 d\phi_2)$ containing $d\varepsilon$ is $-\mathrm{Tr}(d\varepsilon[\phi_1, \phi_2])$, and this cancels the $d\varepsilon$ variation of $\mathrm{Tr}(A[\phi_1, \phi_2])$; the remaining terms cancel by invariance of the trace pairing.

Remark 0.2.8. The same local open model is obtained from the Calabi-Yau 2 point-brane residue calculation by tensoring the skyscraper object with $\mathbb{C}^N$ and passing to endomorphisms. This is a local CY_2 pairing statement, not a global CY-to-chiral input. Global transfer would require the corresponding negative cyclic Calabi-Yau class, framing data, and descent; none of that is used in the proof below.

Remark 0.2.9. In AKSZ language this is the one-dimensional gauged sigma model with Hamiltonian target $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ and symplectic potential $\mathrm{Tr}(\phi_1\, d\phi_2)$. We use the resulting Poisson bivector

$$\Pi = \sum_{i,j} \frac{\partial}{\partial(\phi_1)^i{}_j} \wedge \frac{\partial}{\partial(\phi_2)^j{}_i},$$

so $\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$. Equivalently, with the convention $\iota_{X_f}\omega = -df$ used throughout, the symplectic form is $\mathrm{Tr}(d\phi_1 \wedge d\phi_2)$. For $\epsilon \in \mathfrak{gl}_N$, the Hamiltonian

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big)$$

generates $\delta_\epsilon \phi_i = [\epsilon, \phi_i]$. Its zero locus is the commuting-pair equation $[\phi_1, \phi_2] = 0$, and the classical reduced phase space is the stack of commuting pairs of matrices modulo conjugation.

Remark 0.2.10. Equivalently, take Chern-Simons theory on $\mathbb{R} \times T^2$ and restrict to the harmonic forms $1, \eta_1, \eta_2, \eta_1\eta_2$ on T^2. Writing $A_{3d} = A_t\, dt + \phi_1\eta_1 + \phi_2\eta_2$ and integrating over T^2 gives the same action; the odd variables ε_i represent the degree-one cohomology classes η_i.

0.2.4 Closed mixed Hamiltonian sector

The closed field is the infinitesimal geometry of the formal normal surface. Because the surface is holomorphic symplectic, an infinitesimal canonical transformation is a Hamiltonian function modulo constants. The BF partner is the cotangent variable which makes the theory local and BV. This is the closed counterpart of the open first-order form: both sides are governed by the same symplectic normal plane.

The closed sector used here is a mixed field theory, not a tensor product of two independent closed-string state spaces. Its fields have a topological de Rham factor along $\mathbb{R}^2$ and a holomorphic Dolbeault factor along $\mathbb{C}^2$:

$$\Omega^*(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,*}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big).$$

The coefficient $\widehat{\mathfrak{h}}$ is the completed Lie algebra of nonconstant Hamiltonian functions on the formal holomorphic symplectic disk. The continuous dual copy $\widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[2]$ is the cotangent/BF field. Thus the closed theory is the Hamiltonian Kodaira-Spencer/BF sector of the mixed holomorphic-topological string.

Remark 0.2.11. The original BCOV analysis [1] is written for Calabi-Yau threefolds. On a holomorphic symplectic surface the natural local cotangent theory is the BF theory for the dg Lie algebra of Hamiltonian vector fields. The holomorphic symplectic form supplies the Poisson bracket, while the holomorphic volume form supplies the divergence operator on polyvectors.

We consider the space of polyvector fields

$$\mathbf{PV}^{*,*}(\mathbb{C}^2) \cong \Omega^{0,*}(\mathbb{C}^2)[\partial_{z_1}, \partial_{z_2}]$$

where ∂_{z_i} are odd generators of poly-vector degree 1 (corresponding to the two holomorphic vector fields on $\mathbb{C}^2$). The operator

$$\partial : \mathbf{PV}^{i,*} \to \mathbf{PV}^{i-1,*}$$

is the divergence operator determined by the holomorphic volume form $\Omega = dz_1 \wedge dz_2$:

$$\partial_\Omega = \iota_\Omega^{-1} \circ \partial_{\mathrm{hol}} \circ \iota_\Omega, \qquad \iota_\Omega : \mathbf{PV}^{i,*} \xrightarrow{\sim} \Omega^{2-i,*}.$$

Thus for $X = a_1\partial_{z_1} + a_2\partial_{z_2}$,

$$\partial_\Omega X = (\partial_{z_1} a_1 + \partial_{z_2} a_2),$$

while ∂_Ω vanishes on functions.

We work in the formal polynomial local model at $0 \in \mathbb{C}^2$, so Dolbeault cohomology is represented by holomorphic Taylor polynomials. Here ∂_Ω is the divergence constraint selecting the Hamiltonian BF sector. The kinematic differential of the closed theory below is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$. Thus the Poincare lemma identifies divergence-free vector fields with Hamiltonian potentials. The quotient by ∂_Ω-exact Hamiltonians is absent. Analysing degree by degree:

- $\mathbf{PV}^{0,*}(\mathbb{C}^2) = \Omega^{0,*}(\mathbb{C}^2) \subset \mathbf{Ker}\partial$ trivially.
- $\mathbf{PV}^{1,*}(\mathbb{C}^2) \cong \Omega^{1,*}(\mathbb{C}^2)$ via $\iota_{(-)}\omega$. Under this identification, with the Hamiltonian convention $\iota_{X_f}\omega = -df$ used below, $\mathbf{Ker}\partial_\Omega$ is the space of closed holomorphic one-forms. Lemma 0.2.12 identifies these with df, uniquely modulo constants.
- The constant Hamiltonians act trivially, since their Hamiltonian vector fields vanish. We therefore use the reduced polynomial algebra

  $$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$$

  for nontrivial Hamiltonian fields.

Lemma 0.2.12 *(Polynomial Poincare primitive).* Let

$$\theta = a(z_1, z_2)\, dz_1 + b(z_1, z_2)\, dz_2$$

be a polynomial one-form on $\mathbb{C}^2$. If $\partial_{z_2} a = \partial_{z_1} b$, then there is a polynomial $f \in \mathbb{C}[z_1, z_2]$, unique modulo constants, such that $\theta = df$.

Proof. Define

$$f(z_1, z_2) = \int_0^1 \big(z_1 a(tz_1, tz_2) + z_2 b(tz_1, tz_2)\big)\, dt.$$

Since a and b are polynomials, the integral is termwise division of each homogeneous term by its degree plus one, hence f is polynomial. Differentiate under the integral sign. Closedness says $\partial_{z_2} a = \partial_{z_1} b$, so

$$\begin{aligned}
\partial_{z_1} f &= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_1} b(tz)\big)\, dt \\
&= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_2} a(tz)\big)\, dt \\
&= \int_0^1 \frac{d}{dt}\big(t\, a(tz)\big)\, dt = a(z).
\end{aligned}$$

The same calculation, with a and b interchanged, gives $\partial_{z_2} f = b(z)$. Therefore $df = \theta$. If $df = dg$, then $d(f - g) = 0$, hence $f - g$ is constant. □

PROPOSITION 0.2.13 *(Hamiltonian polyvector reduction).* In the formal polynomial model on $(\mathbb{C}^2, \omega = dz_1 \wedge dz_2)$, the divergence-closed polyvector-one fields reduce to Hamiltonian vector fields:

$$\mathbf{Ker}(\partial_\Omega : \mathbf{PV}^1 \to \mathbf{PV}^0) \cong \{df : f \in \mathbb{C}[z_1, z_2]\} \cong \bar{A}.$$

Under the convention

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df,$$

the commutator of vector fields is

$$[X_f, X_g] = X_{\{f,g\}_{\bar{A}}}.$$

Proof. Let $X = a_1 \partial_{z_1} + a_2 \partial_{z_2}$. The divergence operator is $\partial_\Omega X = \partial_{z_1} a_1 + \partial_{z_2} a_2$. Contracting with ω gives

$$\iota_X \omega = a_1\, dz_2 - a_2\, dz_1.$$

The equation $\partial_\Omega X = 0$ is exactly the closedness of this one-form:

$$d(\iota_X \omega) = (\partial_{z_1} a_1 + \partial_{z_2} a_2)\, dz_1 \wedge dz_2 = 0.$$

By Lemma 0.2.12, there is a polynomial f, unique modulo constants, such that $\iota_X \omega = -df$. Therefore

$$a_1 = -\partial_{z_2} f, \qquad a_2 = \partial_{z_1} f,$$

so $X = X_f$. Constants are exactly the kernel of $f \mapsto X_f$, giving $\bar{A}$.

Finally, for any polynomial h,

$$\begin{aligned}
[X_f, X_g](h) &= \{f, \{g, h\}\} - \{g, \{f, h\}\} \\
&= \{\{f, g\}, h\}
\end{aligned}$$

by the Jacobi identity for the constant Poisson bracket. Thus $[X_f, X_g] = X_{\{f,g\}}$, and projecting constants gives the bracket on $\bar{A}$. □

For the Hamiltonian sector of the closed mixed field theory we replace $\mathbf{Ker}\partial$ (on poly-1-vectors) by the Dolbeault-valued Hamiltonian vector fields:

$$\Omega^{0,*}(\mathbb{C}^2)\text{-valued Hamiltonian vector fields,}$$

i.e. sections of $\Omega^{0,*}(\mathbb{C}^2)$ identified with Hamiltonian vector fields via the symplectic form $\omega = dz_1 \wedge dz_2$. Concretely, a section $f \in \Omega^{0,*}(\mathbb{C}^2)$ corresponds to the Hamiltonian vector field $X_f = \{f, -\}_\omega$, where

$$\{f, g\}_\omega = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g.$$

With this convention

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df.$$

Remark 0.2.14. The replacement is the elementary statement of Proposition 0.2.13. Constants are removed because $X_f = 0$ exactly when f is constant.

0.2.5 Hamiltonian BF closed sector

0.2.5.1 The Action Functional

Let

$$D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

Write

$$\alpha \in \Omega^*(\mathbb{R}^2) \hat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \widehat{\mathfrak{h}}[1]$$

for the Hamiltonian connection field and

$$\beta \in \Omega^*(\mathbb{R}^2) \hat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]$$

for its cotangent field. The bracket on form-valued Hamiltonians is obtained by wedging the form parts and taking the Poisson bracket of the Hamiltonian parts with the tensor-product Koszul sign:

$$[\eta \otimes f, \theta \otimes g] = (-1)^{|f||\theta|}(\eta \wedge \theta) \otimes \{f, g\},$$

for homogeneous form η, θ and homogeneous Hamiltonian coefficients f, g. In the ghost-zero Hamiltonian calculations used below the coefficients are even, so this sign is +1. The action is

$$S = \int_{\mathbb{R}^2 \times \mathbb{C}^2} \left\langle \beta,\ D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle dz_1 \, dz_2.$$

The pairing is the continuous evaluation pairing between $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ and $\widehat{\mathfrak{h}}$. The dense polynomial submodel replaces these by $\bar{A}^\vee$ and $\bar{A}$ in each finite degree. For ghost-number-zero fields, α has total real/Dolbeault form degree 1 and β has total degree 2. Hence $\beta \wedge D\alpha$ and $\beta \wedge \{\alpha, \alpha\}$ have total real/Dolbeault degree 4; multiplying by $dz_1 \wedge dz_2$ gives a top real degree 6 form on $\mathbb{R}^2 \times \mathbb{C}^2$.

Remark 0.2.15. The open mixed brane system lives on the brane line. The closed Hamiltonian BF theory lives on the full mixed space-time $\mathbb{R}^2 \times \mathbb{C}^2$.

0.2.5.2 Classical field content

This theory is closely related to a 4-dimensional BF theory: β plays the role of the B-field (Lagrange multiplier) and the equations of motion of β enforce $F_\alpha = 0$, where

$$F_\alpha = D\alpha + \tfrac{1}{2}\{\alpha, \alpha\}$$

is the "curvature" of α (viewed as a connection on a bundle of holomorphic-symplectic surfaces over $\mathbb{R}^2$).

Consider the space-time $\mathbb{R}^2_{x_i} \times \mathbb{C}^2_{z_i}$. The ghost-number-zero components of α and β are:

$$\alpha = \alpha_{x_i}\, dx_i + \alpha_{\bar{z}_j}\, d\bar{z}_j, \qquad \beta = \beta_{x_1 x_2}\, dx_1\, dx_2 + \beta_{x_i \bar{z}_j}\, dx_i\, d\bar{z}_j + \beta_{\bar{z}_1 \bar{z}_2}\, d\bar{z}_1\, d\bar{z}_2.$$

Here the coefficients of α are Hamiltonian functions modulo constants. The gauge symmetries are the BF-type transformations

$$\delta_\varepsilon \alpha = D\varepsilon + \{\alpha, \varepsilon\}, \qquad \delta_\varepsilon \beta = \mathrm{ad}^\vee_\varepsilon\, \beta,$$

and

$$\delta_\lambda \alpha = 0, \qquad \delta_\lambda \beta = D_\alpha \lambda,$$

where $D_\alpha = D + \{\alpha, -\}$ and λ is a one-form valued in $\bar{A}^\vee$. The second symmetry is reducible on shell: if $\lambda = D_\alpha \mu$, then $D_\alpha \lambda = D_\alpha^2 \mu = \{F_\alpha, \mu\}$, which vanishes when $F_\alpha = 0$.

Analysing α as a Hamiltonian function on $\mathbb{C}^2$: the component $\alpha_{\bar{z}_j} d\bar{z}_j$ is the Hamiltonian potential whose vector field $X_{\alpha_{\bar{z}_j}} d\bar{z}_j$ is the Beltrami differential encoding a deformation of the complex structure of $\mathbb{C}^2$; the component $\alpha_{x_i} dx_i$ similarly defines a connection on $\mathbb{R}^2$ valued in Hamiltonian vector fields on $\mathbb{C}^2$, encoding a bundle of $\mathbb{C}^2$s over $\mathbb{R}^2$. Together, the equation of motion $F_\alpha = 0$ says this data constitutes an integrable deformation of complex structure with flat holonomy. The field β appears linearly in the action and enforces exactly this equation.

Remark 0.2.16. This is entirely analogous to BF theory, where B is a Lagrange multiplier enforcing $F_A = 0$, and the phase space is $T^*\mathcal{M}_{\mathrm{flat}}$.

0.2.5.3 Phase Space

As an optional analytic model for propagators, formally compactify one real direction and put the perturbative local model on $\mathbb{R} \times S^1 \times Y_{\mathrm{hol}}$, where Y_{hol} is holomorphic symplectic and the calculation is completed at the trivial solution. This is a compactified avatar of the formal local model, not a change of notation for the product $X = M \times Y$ used above. The equations of motion $F_\alpha = 0$ cut out the Maurer-Cartan moduli problem for Hamiltonian vector fields on $S^1 \times Y_{\mathrm{hol}}$. Fix the analytic package of Problem 0.2.74 in this compact or formal setting: elliptic complex, brane-defect or boundary condition, gauge fixing, propagator, renormalization scheme, and anomaly cancellation. The perturbative phase space is the derived cotangent stack

$$T^*_{\mathrm{der}}\mathrm{MC}\big(\Omega^*(S^1)\widehat{\otimes}\Omega^{0,*}(Y_{\mathrm{hol}}, \mathrm{Ham}(Y_{\mathrm{hol}})), D, \{-,-\}\big).$$

After passing to the underived tangent space at the trivial solution this reduces to the ordinary cotangent bundle of the linearized deformation complex.

0.2.6 *Closed-open brane coupling*

This is the point at which the physical picture becomes a formula. A closed Hamiltonian is not observed by the brane as an abstract vector field; it is observed by evaluating it on the brane's matrix-valued normal coordinates and taking the trace over Chan-Paton indices.

The local classical bulk-boundary operation is the Hamiltonian shadow of the desired closed-open morphism: evaluate a closed Hamiltonian on the brane normal fields. If the brane lies along x_1 and the closed field is $dx_1\, z_1^k z_2^l$, the local boundary functional is

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big) + \int_{\mathbb{R}} dx_1\, \mathrm{Tr}\, \phi_1^k \phi_2^l.$$

This is a statement inside the formal local algebraic model: the closed Hamiltonian is pulled back along the matrix-valued normal coordinate map $z_i \mapsto \phi_i$ and then traced. Construction 0.2.32 is the algebraic form of this bulk-to-boundary prescription. Theorem 0.2.58 below proves that the resulting Hamiltonian evaluation preserves the classical bracket. This is the degree-zero Hamiltonian component expected from a full closed-open map; the full Poisson factorization-center morphism is not constructed here.

Construction 0.2.17 *(Reduced line-defect current kernel).* Let

$$X = \mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}, \qquad L = \mathbb{R}_{x_1} \times \{x_2 = 0\} \times \{p\} \subset X,$$

and let $i : L \hookrightarrow X$ be the inclusion. This is a line defect, not a spacetime boundary. In the reduced formal local model, a test density $a(t)\, dt$ on L defines the defect-supported current $i_*(a(t)\, dt)$, locally written as $a(x_1)\delta(x_2)\delta_p\, dx_1 dx_2$ together with the formal holomorphic delta-current at p.

The degree-zero Hamiltonian part of the reduced PBW/Rees defect-current vertex is

$$K_L^{\mathrm{cl}}(a \otimes f) = \int_L a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt, \qquad f \in \bar{A}.$$

For the cotangent principal-part source, the reduced vertex sends

$$B \otimes \rho[1] \longmapsto \Theta_\rho(B), \qquad B \in \mathcal{D}'_c(L), \quad \rho \in \mathcal{P},$$

using Proposition 0.2.55. These formulas define a reduced PBW/Rees defect-current kernel. They are not a cochain-level map from closed CE observables, and they are not yet the quantum Costello brane-defect kernel: the Ward identity, anomaly cancellation, finite-scale propagator, and quantum moment-map compatibility remain separate obligations.

Remark 0.2.18. Geometrically, specifying a closed Hamiltonian field on $\mathbb{C}^2$ amounts to equipping $\mathbb{C}^2$ with a connection, so that the brane can parallel-transport its internal degrees of freedom as it moves along $\mathbb{R}$.

For a factorization algebra $\mathcal{A}$ on the line, write $\mathrm{Z}_{\mathrm{fact}}(\mathcal{A})$ for the factorization algebra of E_1 central operations. In the locally constant one-dimensional case its stalk is the Hochschild cochain complex of the corresponding E_1 algebra. The notation $\mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathcal{A})$ means the same central operations with the additional compatibility homotopies for the classical Poisson/BV bracket. The proposition below constructs the ordinary central operations and the induced P_0 bracket identity on stable connected local cohomology. It does not construct all higher P_0 center homotopies.

PROPOSITION 0.2.19 *(Degree-zero Hamiltonian current shadow).* Let $\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}$ be the locally constant replacement of the classical open brane factorization algebra obtained after the de Rham jet contraction of Lemma 0.2.28. Its stalk is the reduced BV algebra $A^{\mathrm{pt}}_N = (\mathcal{R}^{GL_N}_N, Q)$. For an interval $I \subset \mathbb{R}$, put

$$\mathfrak{J}_H(I) = C^\infty_c(I) \otimes \bar{A},$$

with extension by zero for inclusions and with current bracket

$$[a \otimes f, b \otimes g] = ab \otimes \{f, g\}_{\bar{A}}.$$

For $a \in C^\infty_c(I)$ and $f \in \bar{A}$, define the central operation

$$\Phi_{I,N}(a \otimes f)(u) = \left(\int_I a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt\right) u.$$

In the locally constant reduced model this is multiplication by the compactly supported Hamiltonian boundary observable. The assignments $\Phi_{I,N}$ are compatible with inclusions of intervals. Extending multiplicatively from generators gives a prefactorization map from the commutative special fibre $\mathbf{S}(\mathfrak{J}_H)$ to ordinary multiplication operations on $\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}$. After projection to the stable connected Hamiltonian trace quotient, the generator-level local bracket satisfies the P_0 identity

$$\{\Phi(a \otimes f), \Phi(b \otimes g)\}_{\mathrm{open}} = \Phi(ab \otimes \{f, g\}_{\bar{A}}).$$

Its stalk at a point is the boundary map of Theorem 0.2.58.

Proof. In the reduced open model $Q\phi_1 = Q\phi_2 = 0$. Hence $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is Q-closed for every Hamiltonian f, and the compactly supported integral is Q-closed. Since the reduced open observable algebra is graded-commutative, multiplication by this closed element is a Hochschild 0-cocycle; this is the interval-wise version of Proposition 0.2.39. Extension by zero of the test function is compatible with restriction to a larger interval. For disjoint intervals the factorization product of two such operations is multiplication by the product $I_\partial(a, f) I_\partial(b, g)$; this is the multiplicative extension from $\mathfrak{J}_H$ to $\mathbf{S}(\mathfrak{J}_H)$, not the linear current bracket.

The Poisson statement is local on the line. For compactly supported test functions a, b, Corollary 0.2.59 gives

$$\{I_\partial(a, f), I_\partial(b, g)\}_{\mathrm{open}} = I_\partial(ab, \{f, g\}_{\bar{A}}).$$

Passing to the connected stable Hamiltonian quotient removes only the scalar Hamiltonian line, which is central for the bracket. Thus the current bracket is preserved. Taking a and b supported in a small coordinate interval and then passing to the locally constant stalk recovers Theorem 0.2.58. □

PROPOSITION 0.2.20 *(CE-source obstruction).* Let $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$ with the reduced Poisson bracket. Then $\bar{A}$ is a perfect Lie algebra:

$$\bar{A} = \{\bar{A}, \bar{A}\}.$$

If $\xi_{a,b}$ is the CE length-one coordinate dual to the Hamiltonian mode $z_1^a z_2^b$, with $a + b > 0$, then $\xi_{a,b}$ is not closed, and no CE cocycle has CE-length-one part $\xi_{a,b}$. Consequently the boundary assignment

$$f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$$

is a Hamiltonian-mode, or PBW/Koszul-dual, construction. It is not a cochain-level map out of $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ obtained by sending CE linear coordinates dual to Hamiltonian modes to the boundary observables.

Proof. For every monomial $z_1^a z_2^b$ one has, before quotienting constants,

$$z_1^a z_2^b = \left\{ z_1, \frac{1}{b+1} z_1^a z_2^{b+1} \right\}.$$

Hence every nonconstant class in $\bar{A}$ is a Poisson bracket. If $\xi_{a,b}$ denotes the CE linear coordinate dual to the class of $z_1^a z_2^b$, then

$$d_{\mathrm{CE}} \xi_{a,b} \left(z_1, \frac{1}{b+1} z_1^a z_2^{b+1} \right) = -1$$

in the CE convention used here. Thus $\xi_{a,b}$ is not closed. The CE differential raises cochain length by one: if $O = \xi_{a,b} + O_{\geq 2}$, then the CE-length-two part of $d_{\mathrm{CE}} O$ is $d_{\mathrm{CE}} \xi_{a,b}$, and no higher-length term can cancel it. Hence no closed CE cochain has linear Hamiltonian part $\xi_{a,b}$. A cochain-level map from closed CE observables must send closed source cochains to closed central operations. The Hamiltonian boundary observable is closed on the open side by Proposition 0.2.39, but its source is the Hamiltonian mode f, not a closed CE generator. This is the source obstruction. □

Theorem 0.2.21 *(Reduced derived Poisson center).* Let

$$\mathfrak{h} = \prod_{d \geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d \geq 1} H_d^\vee$$

be the product/direct-sum Tate Hamiltonian Lie algebra of the formal symplectic disk and its continuous dual. Put

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

Let $A_\partial = \widehat{\mathbf{S}}(\mathfrak{h})$ be the completed stable degree-zero Hamiltonian trace algebra, with generators O_i corresponding to a homogeneous topological basis H_i of $\mathfrak{h}$. If

$$[H_i, H_j] = C_{ij}^k H_k,$$

with the constant target projected out, equip A_∂ with the linear Poisson bracket

$$\{O_i, O_j\}_\partial = C_{ij}^k O_k.$$

Let

$$\mathrm{PV}_{\mathrm{cont}}(A_\partial) = A_\partial \widehat{\otimes} \widehat{\Lambda}(\mathfrak{h}^\vee_{\mathrm{cont}})$$

be the completed algebra of continuous polyvectors, with odd generators θ_i dual to the coordinates O_i. The Poisson bivector is

$$\pi_\partial = \frac{1}{2} C_{ij}^k O_k \, \theta_i \theta_j,$$

and the Poisson differential is $d_\pi = [\pi_\partial, -]$ in the Schouten convention.

If c_i is the CE coordinate dual to $H_i \in \mathfrak{h}$ and u_i is the CE coordinate dual to the shifted cotangent generator $\delta_i[1] \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$, then the assignment

$$c_i \longmapsto \theta_i, \qquad u_i \longmapsto O_i$$

extends to an isomorphism of completed graded-commutative dg algebras

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} \left(\mathrm{PV}_{\mathrm{cont}}(A_\partial), d_\pi \right).$$

Proof. By the product/direct-sum convention already used in Proposition 0.2.47, the continuous CE algebra of $\mathfrak{g}$ is freely generated, after completion, by the odd coordinates c_i and the even coordinates u_i. The CE differential is dual to the brackets of $\mathfrak{g}$:

$$d_{\mathrm{CE}} c_k = -\frac{1}{2} C_{ij}^k c_i c_j, \qquad d_{\mathrm{CE}} u_k = C_{kj}^{\ell} u_\ell c_j.$$

The second formula is the coadjoint action of $\mathfrak{h}$ on $\mathfrak{h}_{\mathrm{cont}}^\vee[1]$ written on the dual coordinates.

The linear Poisson bivector gives the Lichnerowicz differential [7]

$$d_\pi \theta_k = -\frac{1}{2} C_{ij}^k \theta_i \theta_j, \qquad d_\pi O_k = C_{kj}^{\ell} O_\ell \theta_j.$$

Hence $c_i \mapsto \theta_i$ and $u_i \mapsto O_i$ intertwine the two differentials. The underlying completed graded-commutative algebras are freely generated by corresponding variables, so this is an isomorphism.

Continuity is the same coefficient-finiteness used in Lemma 0.2.46: for each fixed output coefficient in a bracket of Hamiltonians only finitely many homogeneous input-degree pairs contribute, and the coadjoint input has finite support in $\mathfrak{h}_{\mathrm{cont}}^\vee$. Thus the displayed sums define continuous derivations of the completed CE and polyvector algebras. □

Remark 0.2.22. The theorem is the reduced algebraic center calculation. It does not say that the scalar boundary function $O_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is the image of the nonclosed CE coordinate c_f. It says that O_f is the image of the CE coordinate u_f dual to the shifted cotangent generator, while c_f maps to the constant Hamiltonian vector-field operation θ_f. This is why Proposition 0.2.20 remains correct and why the full BV/factorization lift below has nontrivial content.

Construction 0.2.23 *(Compact current PBW/Rees source).* Let $\mathcal{P}$ denote the principal-part module

$$\mathcal{P} = \bigoplus_{\substack{a,b \geq 0 \\ a+b>0}} \mathbb{C}\, \rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

identified with $\mathfrak{h}_{\mathrm{cont}}^\vee$ by the residue pairing of Proposition 0.2.52. Put

$$\mathfrak{g}_{\mathrm{pp}} = \mathfrak{h} \ltimes \mathcal{P}[1],$$

with zero differential and bracket

$$[f, g] = \{f, g\}, \qquad [f, \rho[1]] = (f \cdot \rho)[1], \qquad [\rho[1], \sigma[1]] = 0,$$

where $f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\}$. On monomials this action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1, b-q+1},$$

with negative-index targets zero.

For an interval $I \subset \mathbb{R}$, define the compactly supported current Lie algebra

$$\mathfrak{J}_{\mathfrak{g}}(I) = C_c^\infty(I) \widehat{\otimes} \mathfrak{g}_{\mathrm{pp}}, \qquad [\alpha \otimes x, \beta \otimes y] = \alpha\beta \otimes [x, y].$$

Its PBW/Rees algebra is

$$U_\hbar\big(\mathfrak{J}_{\mathfrak{g}}(I)\big) = \widehat{T}\big(\mathfrak{J}_{\mathfrak{g}}(I)\big)[[\hbar]] \big/ \langle xy - (-1)^{|x||y|} yx - \hbar[x, y] \rangle.$$

The PBW filtration is the tensor-word filtration. Its special fibre is

$$U_\hbar(\mathfrak{J}_\mathfrak{g}(I))/(\hbar) \cong \widehat{\mathbf{S}}(\mathfrak{J}_\mathfrak{g}(I)),$$

equipped with the semiclassical P_0 bracket

$$\{\bar{X}, \bar{Y}\}_{P_0} = \left(\hbar^{-1}(XY - (-1)^{|X||Y|}YX)\right) \bmod \hbar.$$

This is the PBW/Koszul-dual source for Hamiltonian and principal-part modes. It is not a closed CE cochain algebra map; the CE comparison must be made separately by bar-cobar twisting or by a completed pairing.

Problem 0.2.24 *(PBW-source P_0 center action and CE comparison).* Construct a filtered factorization-center action of the compactly supported current PBW/Rees source $U_\hbar(\mathfrak{J}_\mathfrak{g})$ of Construction 0.2.23. The target cannot be the polynomial open observable algebra alone if the cotangent generators are included; it must use the enlarged principal-part boundary object of Proposition 0.2.55. In the classical special fibre the action must send

$$f \longmapsto B_f = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b},$$

and satisfy the stalkwise bracket identities

$$\{B_f, B_g\}_{\text{open}} = B_{\{f,g\}}, \qquad \{B_f, \Theta_\rho\}_{\text{open}} = \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\}_{\text{open}} = 0.$$

This special-fibre statement is the natural P_0 theorem. A strict action of the full filtered Rees algebra on the same commutative reduced target is not a consequence of these identities: in the source

$$XY - (-1)^{|X||Y|}YX = \hbar[X, Y],$$

while multiplication operators on a commutative classical target commute. Thus the Rees-level theorem requires either a deformed boundary target or an explicit homotopical quantization of the central operations. The target data must include differential compatibility, factorization centrality homotopies, coherent P_0 bracket homotopies, and compact-support naturality for interval inclusions and disjoint unions before the de Rham contraction.

Separately compare this PBW/Koszul-dual source with the closed CE observable algebra by an explicit twisting or pairing construction. Without this comparison, the result is a PBW/Koszul-dual boundary action, not a cochain-level map from closed CE observables.

Problem 0.2.25 *(Full BV/factorization lift of the reduced derived center).* Let $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}$ be the classical factorization algebra on the brane line obtained from the cyclic open BV theory

$$\Omega^*(\mathbb{R}) \otimes \mathbf{Ext}^*_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \otimes \mathfrak{gl}_N[1],$$

and let $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}$ be the Hamiltonian cotangent BF factorization algebra on the formal neighbourhood of $\mathbb{R} \times p$. Since the classical open stalk is graded-commutative, ordinary E_1 centrality of multiplication is not the main theorem. The target must retain the classical Poisson/BV bracket.

Lift Theorem 0.2.21, already in the formal local classical model, to a cochain-level morphism of factorization algebras

$$\Phi_{\mathrm{cl}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_{\mathbb{R}\times p} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N})$$

extending Proposition 0.2.19 and whose stable connected Hamiltonian local cohomology has the following restrictions:

(i) the source convention is fixed by Theorem 0.2.21: the boundary Hamiltonian function is the image of the cotangent CE coordinate u_f, while the Hamiltonian CE coordinate c_f maps to the corresponding constant vector-field operation. The lift must realize these operations inside the actual factorization center, not only in the reduced polyvector model;

(ii) this source issue is a real obstruction, not notation: Proposition 0.2.20 proves that the reduced Hamiltonian Poisson algebra is perfect, so the CE linear coordinate dual to a Hamiltonian mode is not closed;

(iii) for the cotangent coordinate u_f, the induced stalk operation is multiplication by the boundary observable $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$, whose Hochschild-central stalk cocycle is Proposition 0.2.39;

(iv) the shifted cotangent generator cannot be realized by the polynomial descendants $\Psi_{k,l}$ as a deformation-level coadjoint action: Corollary 0.2.51 shows the mismatch. A full cotangent lift must construct a Laurent, distributional, or other boundary model for $\mathfrak{h}^\vee_{\mathrm{cont}}$. The natural algebraic target is the principal-part module of Proposition 0.2.52. Boundary operations $\Theta_{a,b}$ for those modes would replace the polynomial $\Psi_{a,b}$ in the cotangent lift. Proposition 0.2.55 constructs this replacement only in the reduced post-contraction current model; the unreduced BV/factorization homotopies are still part of the present problem;

(v) the factorization centrality homotopies identify insertion of $\Phi_{\mathrm{cl}}(f)$ to the left or right of any open local observable, with the Koszul sign determined by cohomological degree;

(vi) the induced Poisson bracket on local cohomology restricts on degree-zero Hamiltonians to Theorem 0.2.58.

This is stronger than the boundary bracket theorem and stronger than ordinary central multiplication. Proposition 0.2.19 constructs the degree-zero current shadow: ordinary central operations and the induced P_0 bracket identity on stable connected local cohomology. The full problem asks in addition for compact-support naturality before the de Rham contraction, an HKR/polyvector model with the same Tate topology as Theorem 0.2.21, refinement homotopies for the closed CE observables, all Poisson/BV center homotopies, the stable connected quotient, and a boundary realization of the shifted cotangent generators. Any global version must additionally pass through the local-Hamiltonian descent and period obstruction of Proposition 0.1.13; a stalkwise formal disk map is not a Weiss/Ran descent theorem for factorization centers.

Remark 0.2.26 *(Weiss/Ran descent firewall).* A global closed-open map is not obtained by choosing Darboux coordinates at each point and gluing the resulting stalk maps. The obstruction lives in the Weiss/Ran total complex

$$\mathrm{Tot}\, C^\bullet_{\mathrm{WR}}\Big(\mathrm{Ran}(L); \underline{\mathrm{RHom}}\Big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}, Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})\Big)\Big).$$

Vanishing of the local Hamiltonian period class is only the first condition. One must also prove source comparison, cotangent boundary realization, all P_0-center homotopies, and factorization descent. Until then the Hamiltonian boundary map, the Θ-currents, and the PBW/Rees action are formal local or reduced-current statements.

0.2.7 Operators in the open brane gauge theory and the large N limit

The local-operator calculation is performed in the reduced BV model with fields ϕ_1, ϕ_2 and the antifield $\psi = A^*$ for the Lagrange multiplier A. We take GL_N-invariants in characteristic zero, so the Chevalley part of the gauge BRST differential acts trivially on trace functions by cyclicity.

The gauge field has one algebraic job: it imposes the moment-map equation $[\phi_1, \phi_2] = 0$. In BV language this equation is imposed by the Koszul differential $Q\psi = [\phi_1, \phi_2]$. This is why the stable trace algebra starts noncommutative, as cyclic words in two matrices, and then collapses in cohomology to commutative monomials on the formal symplectic plane.

Construction 0.2.27 *(The reduced BV algebra at the insertion point).* At a fixed point of the brane line, introduce three matrix variables

$$\phi_1, \phi_2 \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1].$$

The reduced local polynomial algebra is

$$\mathcal{R}_N = \mathbb{C}[(\phi_1)^i{}_j, (\phi_2)^i{}_j] \otimes \Lambda\big((\psi)^i{}_j\big).$$

Its BRST differential is the derivation determined on generators by

$$Q(\phi_1)^i{}_j = 0, \qquad Q(\phi_2)^i{}_j = 0, \qquad Q(\psi)^i{}_j = ([\phi_1, \phi_2])^i{}_j.$$

This formula is the algebraic shadow of the A-equation of motion: varying

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives $[\phi_1, \phi_2] = 0$, and the antifield $\psi = A^*$ records that equation as a Koszul differential.

Lemma 0.2.28 *(De Rham jets).* Positive jets in the topological direction vanish in local operator cohomology. Equivalently, every cohomology class has a representative determined by the values of the fields and antifields at the insertion point.

Proof. For a coordinate t on the line, the de Rham part of the BRST operator is $d_{\mathbb{R}}$. On the jet algebra one has the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$. Hence a local expression containing a positive t-derivative is cohomologous to zero after applying the usual Euler homotopy on jet degree. This kills topological-direction jets and leaves the holomorphic z_i-derivatives used later on the closed side. □

Proposition 0.2.29 *(Stable trace invariants).* Let $F = \mathbb{C}\langle x_1, x_2\rangle$ and let $\mathrm{Cyc}(F) = F/[F, F]$ be the vector space of cyclic words. For each N, GL_N acts on $\mathfrak{gl}_N^2$ by simultaneous conjugation. The degreewise stable invariant algebra

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} := \bigoplus_{d\geq 0} \varprojlim_N \big(\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}\big)_d$$

with respect to block-restriction maps in each word-degree is the free commutative algebra generated by the nonempty cyclic words:

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big),$$

where a cyclic word w maps to the trace function $(\phi_1, \phi_2) \mapsto \mathrm{Tr}\, w(\phi_1, \phi_2)$. More precisely, in total positive word-degree d this map is an isomorphism for every $N \geq d$, after omitting the scalar empty trace as a primitive generator.

Proof. Procesi–Razmyslov invariant theory [18, 19] gives two inputs. First, $\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}$ is generated by trace functions of words in the two generic matrices. Second, all polynomial relations among these trace functions are consequences of the fundamental trace identities of matrix size N; in particular every homogeneous trace relation has total word-degree at least $N+1$. Fix a total word-degree d. For every $N \geq d$ the degree-d relations are exactly cyclicity of the trace. Passing degree by degree to the stable inverse limit removes all finite-N trace identities. The surviving identification is $\mathrm{Tr}(uv) = \mathrm{Tr}(vu)$. This is exactly the symmetric algebra on cyclic words.

□

LEMMA 0.2.30 *(Stable marked trace invariants).* In the stable invariant theory used below, the primitive part linear or quadratic in an odd marked matrix ψ is obtained from the corresponding even three-matrix trace invariant by polarization and skew-symmetrisation in the marked letters. The only stable single-trace relation is graded cyclicity: moving one ψ through the trace has sign +1, while moving one ψ past another ψ gives a minus sign.

Proof. Apply Proposition 0.2.29 to three even generic matrices and take the homogeneous summand linear, respectively quadratic, in the third matrix. The Procesi–Razmyslov trace identities have no stable relations in the fixed total word-degree once N is large enough. Declaring the marked matrix odd is the usual super-polarization: the linear part is unchanged, and the quadratic part is projected to the alternating component under exchange of the two marked occurrences. The trace relation therefore becomes exactly graded cyclicity. These are the signs used in the finite complex for one- and two-ψ words. □

Remark 0.2.31. The empty trace is the finite-rank scalar $\mathrm{Tr}_N(1) = N$, which changes with N and fails to define a stable primitive generator. The stable symmetric algebra still has its ordinary unit; what is omitted is the empty trace as a generator of the connected primitive part. The class $\mathrm{Tr}(\psi)$ is a genuine stable odd primitive class. It is excluded from the Hamiltonian comparison because constants have been removed from the closed Hamiltonian algebra.

Varying

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives the moment-map equation $[\phi_1, \phi_2] = 0$. In the reduced BV complex,

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0$$

on invariant trace functions.

Construction 0.2.32 *(Boundary evaluation of a closed Hamiltonian).* Let $f = \sum_{k,l} c_{k,l} z_1^k z_2^l \in \mathbb{C}[z_1, z_2]$. The brane replaces the two normal coordinates z_1, z_2 by the two adjoint scalar fields ϕ_1, ϕ_2. Choosing the ordered lift with all ϕ_1's to the left of all ϕ_2's gives the invariant function

$$\widetilde{\delta}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l}\, \mathrm{Tr}(\phi_1^k \phi_2^l) \in \mathcal{R}_N^{GL_N}.$$

Let

$$H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q) := H^0(\mathcal{R}_N^{GL_N}, Q)/\mathbb{C} \cdot \mathrm{Tr}(1)$$

as a vector space, and as a Lie algebra for the Poisson bracket. This is not an associative quotient of the operator algebra by its unit. Constants give the scalar $c_{0,0}N$ and therefore vanish in this Hamiltonian quotient. Thus the construction descends to a linear map

$$\partial_{\mathrm{bb},N} : \bar{A} \to H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q), \qquad [f] \longmapsto \overline{\widetilde{\partial}_{\mathrm{bb},N}(f)}.$$

Let $\mathsf{W}_{k,l}$ be the set of words in two letters containing k copies of the first letter and l copies of the second. The first descendant is obtained by inserting one Koszul variable and summing over all ordered lifts: For $k + l > 0$ the descendant part of the Hamiltonian comparison is

$$\partial^{(1)}_{\mathrm{bb},N}(\delta_{k,l}[1]) = \Psi_{k,l}, \qquad \Psi_{k,l} := \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big(\psi\, w(\phi_1, \phi_2)\big).$$

The same formula at $(k, l) = (0, 0)$ gives the decoupled scalar antifield $\mathrm{Tr}(\psi)$, which is excluded from the Hamiltonian pairing because constants have been removed from $\bar{A}$.

Lemma 0.2.33 *(First descendants are cycles).* For every $k, l \geq 0$, the symmetrised trace $\Psi_{k,l}$ satisfies $Q\Psi_{k,l} = 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Since x and y are even and $Q\psi = [x, y]$, one has

$$Q\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big((xy - yx)w(x, y)\big).$$

Group the resulting traces by cyclic word of multidegree $(k + 1, l + 1)$. The coefficient of a cyclic word in the positive sum is the number of cyclic adjacent occurrences of the block xy; the coefficient in the negative sum is the number of cyclic adjacent occurrences of the block yx. On a circle the number of transitions from an x-run to a y-run equals the number of transitions from a y-run to an x-run. The two grouped sums are therefore equal, so $Q\Psi_{k,l} = 0$. □

Proposition 0.2.34 *(Primitive one-antifield Koszul homology).* Fix $k, l \geq 0$. The middle homology of the primitive single-trace complex with one displayed ψ and k, l displayed x, y letters is one-dimensional, generated by the class of $\Psi_{k,l}$. The nonconstant part relevant to the Hamiltonian comparison is the direct sum over $k + l > 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Let $\mathsf{W}_{k,l}$ be the set of ordinary words with k letters x and l letters y, and let $\mathsf{N}_{a,b}$ be the set of cyclic words with a letters x and b letters y. By Lemma 0.2.30, primitive cyclic words are taken modulo the graded cyclic rule

$$\mathrm{Tr}(ab) = (-1)^{|a||b|}\mathrm{Tr}(ba), \qquad |x| = |y| = 0, \quad |\psi| = 1.$$

Graded cyclicity gives a unique representative $\mathrm{Tr}(\psi w)$ for every one-ψ cyclic word. Thus the relevant one-ψ chain group is

$$C_1(k, l) = \mathbb{C}\{\mathsf{W}_{k,l}\}.$$

The zero-ψ target is

$$C_0(k, l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\},$$

and the differential is the incidence map

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}.$$

The two-ψ source is generated by traces $\mathrm{Tr}(\psi u \psi v)$ with u and v ordinary words whose total bidegree is $(k-1, l-1)$; if $k = 0$ or $l = 0$ this source is zero. Since ψ is odd, exchanging the two displayed ψ's changes sign. Applying $Q\psi = xy - yx$ gives

$$\begin{aligned}\partial_2(u, v) &= Q\,\mathrm{Tr}(\psi u \psi v)\\ &= \mathrm{Tr}\big(\psi vxyu - \psi vyxu - \psi uxyv + \psi uyxv\big),\end{aligned}$$

so, after removing the common front ψ,

$$\partial_2(u, v) = vxyu - vyxu - uxyv + uyxv.$$

A direct substitution shows $\partial_1\partial_2 = 0$.

If $k = 0$ or $l = 0$, then $C_1(k, l)$ is spanned by x^k or y^l, $\partial_1 = 0$, and $C_2(k, l) = 0$. Hence the middle homology is already the line spanned by $\Psi_{k,l}$. Assume from now on that $k, l > 0$.

We compute the first homology of this finite complex. Form a graph $\Gamma_{k,l}$ whose vertices are the necklaces in $\mathsf{N}_{k+1,l+1}$ and whose oriented edges are the words $w \in \mathsf{W}_{k,l}$, with

$$w : [yxw]_{\mathrm{cyc}} \longrightarrow [xyw]_{\mathrm{cyc}}.$$

Then ∂_1 is the cellular boundary of this graph. For each pair (u, v) as above attach the square whose oriented boundary is $vxyu - vyxu - uxyv + uyxv$. Denote the resulting two-complex by $X_{k,l}$.

Equivalently, $X_{k,l}$ is the two-skeleton of the chip-moving state complex $Y_{k,l}$. The full complex $Y_{k,l}$ has an r-cell for r mutually chosen chip moves on a cycle with $l + 1$ oriented edges and $k + 1$ indistinguishable chips, modulo cyclic relabelling of the edge labels. A vertex is a cyclic distribution of the chips among the vertices of this cycle, hence a necklace in $\mathsf{N}_{k+1,l+1}$. A one-cell moves one chosen chip across one oriented edge; cutting the cycle at this moving edge records exactly a word $w \in \mathsf{W}_{k,l}$, and its two endpoints are $[yxw]_{\mathrm{cyc}}$ and $[xyw]_{\mathrm{cyc}}$. A two-cell records two chip moves performed independently, and its oriented boundary is precisely $vxyu - vyxu - uxyv + uyxv$. Therefore the cellular chain complex of $Y_{k,l}$ through degree two is exactly

$$C_2(k, l) \longrightarrow C_1(k, l) \longrightarrow C_0(k, l),$$

and the inclusion $X_{k,l} = Y_{k,l}^{(2)} \hookrightarrow Y_{k,l}$ induces an isomorphism on H_1, since cells of dimension at least three do not affect first homology.

The homology of $Y_{k,l}$ is the homology of the cycle. For completeness we recall the elementary Abel-map argument. Before quotienting by cyclic relabelling, send a chip configuration to its Abel coordinate, the sum of the chip positions in $\mathbb{R}/(l + 1)\mathbb{Z}$, and extend linearly over each cell. In the universal cover of the target circle, a fibre is described by weakly ordered lifted chip positions with fixed total sum; it is a convex polytope, subdivided by the same chip-crossing cells. Hence the Abel map is a cellular homotopy equivalence to S^1. The finite cyclic relabelling of the y-gaps is equivariant for this Abel map: it rotates the target circle, and every stabilizer acts affinely on a convex fibre, whose quotient is again contractible. Thus the quotient target is still a circle and the quotient complex $Y_{k,l}$ has the homology of S^1. Consequently

$$H_1(X_{k,l}; \mathbb{C}) \cong H_1(S^1; \mathbb{C}) \cong \mathbb{C}.$$

Lemma 0.2.33 identifies this representative as a cycle in the original trace complex. Its non-boundary property follows from the functional

$$\epsilon\big(\mathrm{Tr}(\psi w)\big) = 1, \qquad w \in \mathsf{W}_{k,l},$$

kills every displayed two-ψ boundary because each boundary has two positive and two negative summands, while

$$\epsilon(\Psi_{k,l}) = \binom{k+l}{k} \neq 0.$$

Thus the one-ψ homology in bidegree (k, l) is the one-dimensional line generated by $[\Psi_{k,l}]$. □

Remark 0.2.35. The script `scripts/check_one_psi_homology.py` constructs the finite complex $C_2(k, l) \to C_1(k, l) \to C_0(k, l)$ over $\mathbb{Q}$ and verifies $\mathbf{dim}(\ker)\partial_1/\operatorname{im}\partial_2 = 1$ for $0 \leq k, l \leq 5$. It also checks the open Hamiltonian action of Proposition 0.2.50 for exponents at most 3, and checks the closed coadjoint Taylor-dual formula and the principal-part realization of Proposition 0.2.52 for exponents at most 5. These checks verify the displayed finite models without adding assumptions.

Corollary 0.2.36 *(Boundary pairing for cotangent generators).* Let $\delta_{k,l} \in \mathfrak{h}^\vee$ be the functional dual to the monomial $z_1^k z_2^l \in \bar{A}$. The assignment

$$\delta_{k,l}[1] \longmapsto [\Psi_{k,l}]$$

extends to an isomorphism from the shifted Taylor-dual space $\mathfrak{h}^\vee[1]$ to the nonconstant primitive one-ψ brane homology.

Proof. The monomials $z_1^k z_2^l$ with $k + l > 0$ form the polynomial basis of $\bar{A}$ and the topological basis of $\widehat{\mathfrak{h}}$; the $\delta_{k,l}$ form the continuous coefficient-dual basis. By Proposition 0.2.34, the primitive one-ψ homology has one generator in each bidegree, namely $[\Psi_{k,l}]$. Discarding the scalar bidegree $(0, 0)$ and sending the dual basis vector in every nonconstant bidegree to that generator gives a filtered vector-space isomorphism. This is the boundary pairing used in the PBW special fibre; the closed module structure remains coadjoint. □

Lemma 0.2.37 *(Adjacent swaps).* In the invariant reduced BV cohomology,

$$\operatorname{Tr}(u\phi_1\phi_2 v) - \operatorname{Tr}(u\phi_2\phi_1 v)$$

is Q-exact for all words u, v.

Proof. By cyclicity of the trace,

$$\begin{aligned} Q\operatorname{Tr}(\psi vu) &= \operatorname{Tr}([\phi_1, \phi_2]vu) \\ &= \operatorname{Tr}(\phi_1\phi_2 vu - \phi_2\phi_1 vu) \\ &= \operatorname{Tr}(u\phi_1\phi_2 v) - \operatorname{Tr}(u\phi_2\phi_1 v). \end{aligned}$$

□

Corollary 0.2.38 *(Matrix evaluation is commutative in cohomology).* If two words in ϕ_1, ϕ_2 have the same numbers of ϕ_1 and ϕ_2, then their traces define the same class in the reduced BV cohomology. Consequently $\partial_{\mathrm{bb},N}(f)$ is determined by the commutative polynomial $f \in \mathbb{C}[z_1, z_2]$ and is independent of the chosen ordered lift.

Proof. Adjacent transpositions generate all reorderings of a finite word. Each adjacent transposition of the adjacent letters ϕ_1, ϕ_2 changes the trace by a Q-exact term by Lemma 0.2.37; cyclicity moves the chosen adjacent pair into the displayed position of that lemma. Summing the resulting exact differences proves the claim. □

PROPOSITION 0.2.39 *(Stalk central multiplication).* Let

$$A_N^{\mathrm{pt}} = (\mathcal{R}_N^{GL_N}, Q)$$

be the invariant reduced open BV algebra at a brane insertion point. For $f \in \mathbb{C}[z_1, z_2]$, set

$$B_f = \widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l} \mathrm{Tr}(\phi_1^k \phi_2^l).$$

Then $QB_f = 0$. Regard B_f as a Hochschild 0-cochain; its associated central operation is multiplication

$$L_{B_f} : A_N^{\mathrm{pt}} \to A_N^{\mathrm{pt}}, \qquad a \longmapsto B_f a.$$

This Hochschild 0-cochain is closed. The same statement holds for multiplication by a closed one-antifield class $\Psi_{k,l}$. After projection to the Hamiltonian quotient and stabilization in N, the degree-zero classes satisfy

$$\{B_f, B_g\}_{\mathrm{open}} = B_{\{f,g\}_{\bar{A}}}.$$

Proof. Since $Q\phi_1 = Q\phi_2 = 0$, every B_f is Q-closed. The algebra A_N^{pt} is graded-commutative, so the Hochschild differential of a 0-cochain b is

$$d_{\mathrm{Hoch}} b(a) = Qb\, a + ba - (-1)^{|b||a|} ab.$$

This vanishes for every closed graded-central element b. The classes $\Psi_{k,l}$ are closed by Lemma 0.2.33 and are central for the same graded-commutative reason. Ordered-lift independence for B_f is cohomological and is exactly Corollary 0.2.38. The Poisson identity is Theorem 0.2.58. □

Let $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The removed constant corresponds on the gauge side to the scalar $\mathrm{Tr}(1) = N$ and has zero Hamiltonian vector field.

PROPOSITION 0.2.40 *(Connected brane Hamiltonian sector).* In the stable large-N open BV model, the connected Hamiltonian descendant sector freely generated by the nonempty primitive ghost-zero traces and the nonconstant primitive one-ψ Koszul homology is

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H} \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]\big).$$

The class of $z_1^k z_2^l$ with $k + l > 0$ corresponds to $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$. The corresponding descendant basis vector corresponds to the symmetrised trace with one ψ and k, l copies of ϕ_1, ϕ_2.

Proof. Proposition 0.2.29 gives the free commutative algebra on cyclic words. Lemma 0.2.37 identifies any two cyclic words with the same numbers of ϕ_1 and ϕ_2, because adjacent transpositions generate all reorderings. Thus the connected ghost-zero cyclic words reduce to the monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ with $k + l > 0$. Proposition 0.2.34 computes the primitive one-ψ homology in each multidegree; the nonconstant generators are $\Psi_{k,l}$ with $k + l > 0$. Products of the connected ghost-zero primitive classes and these one-ψ primitive classes give the free graded-commutative multi-trace algebra in the Hamiltonian sector. By definition of the Hamiltonian sector, single traces containing two or more occurrences of ψ are excluded from this algebra. □

Remark 0.2.41. The word "sector" is literal. The calculation above presents the stable algebra generated by the primitive no-ψ Hamiltonians and the nonconstant primitive one-ψ homology generated by $\Psi_{k,l}$ for $k + l > 0$. The full invariant Koszul homology in higher ψ-count lies outside this sector, as does the decoupled scalar class $\mathrm{Tr}(\psi)$.

Problem 0.2.42 *(Full primitive Koszul homology).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| = 1,$$

with differential $d\psi = xy - yx$ and $dx = dy = 0$. Put

$$\mathrm{Cyc}(A_\psi) = A_\psi/[A_\psi, A_\psi]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

Give x, y, ψ weights

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For fixed (R, S) define

$$K^p_{R,S} = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $0 \le p \le \min(R, S)$. The differential is

$$\partial[w] = \sum_{i:a_i=\psi} (-1)^{\#\{\psi \text{ before } i\}} [a_1 \cdots a_{i-1} x y\, a_{i+1} \cdots a_n \\ - a_1 \cdots a_{i-1} y x\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

Prove the primitive homology formula

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \ge 1, \\ 0, & p \ge 2. \end{cases}$$

Equivalently, prove that the primitive stable Koszul homology is $\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta$ with $|\theta| = 1$ and $\mathrm{wt}(\theta) = (1, 1)$.

The required new ingredient is the all-p orientation statement identifying this graded cyclic complex with the cellular chains of the chip-moving necklace state complex, together with the Abel-map proof that the relevant state complex has the homology of S^1. Proposition 0.2.34 proves exactly the $p \le 2$ skeleton needed for the Hamiltonian sector. The complete theorem also requires the all-ψ stable trace presentation and an exact computational check of the finite complex.

0.2.8 Operators in the closed Hamiltonian sector

Let $\mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ with the projected Poisson bracket, and let $\mathfrak{h}^\vee$ be its continuous Taylor dual. The cotangent field is a coadjoint module, so the local Hamiltonian Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

For $f, g \in \mathfrak{h}$ and $\lambda, \mu \in \mathfrak{h}^\vee$, the brackets are

$$[f, g] = \{f, g\}, \qquad [f, \eta_\lambda] = \eta_{\mathrm{ad}^\vee_f \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0,$$

where $(\mathrm{ad}_f^\vee \lambda)(g) = -\lambda(\{f, g\})$. On polynomial monomials in the dense subalgebra $\bar{A}$ this reads

$$[z_1^k z_2^l, z_1^m z_2^n] = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

where the bar means projection to $\bar{A}$; in particular the constant bracket of z_1 and z_2 is zero in the reduced Hamiltonian Lie algebra.

LEMMA 0.2.43 *(Shifted coadjoint signs).* Let η_λ denote the suspension of $\lambda \in \mathfrak{h}^\vee$ into $\mathfrak{h}^\vee[1]$. Since the Hamiltonians in $\mathfrak{h}$ are even,

$$[f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0.$$

Consequently the enveloping algebra relations involving two shifted coadjoint generators are odd-commutative:

$$\eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda = 0.$$

Proof. The semidirect product bracket before shifting is the ordinary coadjoint action of the even Lie algebra $\mathfrak{h}$ on its continuous dual. Suspending the module raises the degree of every λ by one. The mixed-bracket sign is $+1$ because $|f| = 0$. The shifted module is abelian as a Lie ideal, so its bracket with itself is zero; in the associative enveloping algebra this zero Lie bracket gives $\eta_\lambda \eta_\mu - (-1)^{1\cdot 1} \eta_\mu \eta_\lambda = 0$. □

The kinematic BRST operator is

$$Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

The same jet contraction as in Lemma 0.2.28, together with the Dolbeault Poincare lemma in the formal local model, leaves precisely the holomorphic Taylor coefficients. Thus the linear local coordinates are the functionals

$$\partial_{z_1}^k \partial_{z_2}^l \alpha|_0, \qquad \partial_{z_1}^k \partial_{z_2}^l \beta|_0.$$

With the coefficient-dual convention fixed above, these are represented by $(k!l!)^{-1} \partial_{z_1}^k \partial_{z_2}^l|_0$. Since zero-form components of α and β have negative field ghost numbers, the corresponding linear functions have positive ghost numbers.

PROPOSITION 0.2.44 *(Closed local operators are continuous CE cochains).* The classical local operators of the formal Hamiltonian cotangent BF sector are the continuous Chevalley-Eilenberg cochains

$$\mathcal{O}_{\mathrm{grav}}^{\mathrm{cl}} \cong C_{\mathrm{CE,cont}}^\bullet(\mathfrak{g}) = \widehat{\mathbf{S}}(\mathfrak{g}_{\mathrm{cont}}^\vee[-1]),$$

with CE differential dual to the Lie brackets above.

Proof. Work at the formal insertion point. The jet algebra of the fields is the completed free graded-commutative algebra on the Taylor coefficients of α and β. The kinematic differential $Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ contracts all positive real and anti-holomorphic jets: in the real directions the homotopy is $[d_{\mathbb{R}^2}, \iota_{\partial_{x_i}}] = \partial_{x_i}$, and in the anti-holomorphic directions it is the formal Dolbeault Poincare homotopy. The Q_0-cohomology of the jet algebra is therefore generated by the holomorphic Taylor coefficients alone.

The remaining α-coefficients are Taylor Hamiltonians modulo constants, hence elements of $\mathfrak{h} = \widehat{\bar{\mathfrak{h}}}$; degreewise polynomial coefficients form the dense subalgebra $\bar{A}$. The cotangent field β supplies the odd shifted continuous dual $\mathfrak{h}^\vee[1]$. Thus the linear space of residual local modes is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$.

The nonlinear BRST differential is determined by the BV Hamiltonian

$$S = \int \left\langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle.$$

The linear part is exactly Q_0, already killed in cohomology. The quadratic term sends two Hamiltonian α-modes f, g to their reduced Poisson bracket $\{f, g\}$, and the action on the cotangent variable is the coadjoint action displayed above. The quadratic term has zero component with two cotangent inputs, so $[\eta_\lambda, \eta_\mu] = 0$. Hence the residual BRST differential on functions is precisely the derivation dual to the Lie bracket on $\mathfrak{g}$, which is the Chevalley-Eilenberg differential. Continuity records the completion over polynomial degree. □

0.2.9 *Koszul Duality*

Definition 0.2.45. For an augmented complete dg algebra B, its Koszul dual is

$$B^! = \mathrm{RHom}_B(\mathbb{C}, \mathbb{C}).$$

We regard commutative Chevalley-Eilenberg cochains as E_1 algebras by restriction. All bar and cobar constructions below are taken in the category of complete filtered Tate vector spaces, with continuous maps. The completions are by symmetric or tensor length together with the coefficient topology coming from Taylor degree. For a Tate space V, the notation $V^\vee_{\mathrm{cont}}$ means the continuous dual with its dual Tate topology, $\widehat{\otimes}$ means the completed Tate tensor product, and

$$\widehat{\mathbf{S}}^n(V) = (V^{\widehat{\otimes} n})_{\mathbb{S}_n}.$$

Chevalley-Eilenberg chains below are direct sums in CE length with completed coefficients in each length; CE cochains, completed cobar algebras, and completed enveloping algebras are products in the corresponding length filtration.

Lemma 0.2.46 *(Tate CE chains, completed PBW, and Koszul duality).* Let

$$L = \left(\prod_{d\geq 1} P_d\right) \oplus \left(\bigoplus_{d\geq 1} Q_d\right)$$

be a graded Tate vector space over $\mathbb{C}$, with all P_d, Q_d finite dimensional and with chosen topological coefficient bases. Give $\prod P_d$ the product topology and $\bigoplus Q_d$ the discrete direct-sum topology. Assume that L is a graded Lie algebra whose bracket is continuous and coefficient-finite for this product/direct-sum topology:

(i) for the product-product part, every fixed output coefficient receives contributions from only finitely many input degree pairs;

(ii) for every finite direct-sum support $S \subset \mathbb{Z}_{\geq 1}$ in a Q input, the bracket with an arbitrary product input depends on only finitely many product degrees and lands in a finite direct sum of Q degrees determined by S;

(iii) the same finite-window condition holds after restricting both inputs to finite sets of degrees.

Equivalently, every coefficient of the bracket on completed tensors is a finite sum once the direct-sum inputs have their prescribed finite support. Then

$$L^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} P_d^\vee \oplus \prod_{d\geq 1} Q_d^\vee,$$

the natural map $L \to L^{\vee\vee}_{\mathrm{cont}}$ is an isomorphism in this Tate category. Define

$$C^{\mathrm{CE}}_*(L) = \bigoplus_{n\geq 0} \widehat{\mathbf{S}}^n(L[1]),$$

where only the coefficient topology inside each symmetric power is completed. Its continuous dual is

$$C^{\bullet}_{\mathrm{CE,cont}}(L) = \mathrm{Hom}_{\mathrm{cont}}\big(C^{\mathrm{CE}}_*(L), \mathbb{C}\big) = \prod_{n\geq 0} \mathrm{Hom}_{\mathrm{cont}}\big(\widehat{\mathbf{S}}^n(L[1]), \mathbb{C}\big) = \widehat{\mathbf{S}}\big(L^{\vee}_{\mathrm{cont}}[-1]\big).$$

With completed tensor and symmetric algebras defined by tensor length and the coefficient topology, the canonical twisting map induces a filtered quasi-isomorphism

$$\widehat{\Omega}\, C^{\mathrm{CE}}_*(L) \xrightarrow{\sim} \widehat{U}(L),$$

and

$$\mathrm{gr}_{\mathrm{PBW}}\, \widehat{U}(L) \cong \widehat{\mathbf{S}}(L).$$

Proof. A continuous functional on $\prod P_d$ depends on finitely many homogeneous coordinates. A continuous functional on the discrete direct sum $\bigoplus Q_d$ is an arbitrary product of degreewise functionals. This proves the displayed continuous-dual formula and the corresponding Tate reflexivity; it is the product/direct-sum Tate convention of Beilinson–Feigin–Mazur [20].

The coefficient topology is not implemented by truncating Taylor degree as a Lie quotient. In the Hamiltonian application the Poisson bracket lowers Hamiltonian degree, and the coadjoint action can raise dual degree as a formal degree rule. The relevant finiteness is instead the support condition in the direct-sum dual input: every actual dual input has finite Taylor support. The Chevalley-Eilenberg coalgebra is therefore not completed in symmetric length. It is the conilpotent coalgebra

$$C^{\mathrm{CE}}_*(L) = \bigoplus_{n\geq 0} \widehat{\mathbf{S}}^n(L[1]),$$

with completed coefficients in each length. The bracket defines the usual CE coderivation, which lowers symmetric length by one. The coefficient-finiteness hypothesis makes each output coefficient a finite sum, hence the coderivation is continuous. Dualizing the direct sum over n gives the completed product over cochain length, namely $\widehat{\mathbf{S}}(L^{\vee}_{\mathrm{cont}}[-1])$.

Filter $T(L)$ by word length and complete. The closed enveloping algebra is the quotient of $\widehat{T}(L)$ by the closed ideal generated by

$$xy - (-1)^{|x||y|} yx - [x, y].$$

Choose an ordered topological coefficient basis compatible with finite degree windows. PBW straightening replaces one inversion by the ordered word plus one bracket term. Coefficient-finiteness implies that every coefficient of every straightened completed word is computed by finitely many reductions. Thus ordinary graded PBW, applied on finite-dimensional degree windows and then passed to the completion, gives

$$\mathrm{gr}_{\mathrm{PBW}}\, \widehat{U}(L) \cong \widehat{\mathbf{S}}(L)$$

[21, 22].

The cobar comparison is checked with the same PBW/order filtration. The canonical Lie twisting cochain

$$C^{\mathrm{CE}}_*(L) \longrightarrow \widehat{U}(L)$$

sends a suspended cogenerator to the corresponding enveloping generator. Its associated graded map is the completed Koszul complex of $\widehat{\mathbf{S}}(L)$. The usual Euler contracting homotopy on the Koszul complex is $h = \iota_E/N$ in positive total Koszul degree, where E is the Euler vector field and N is total polynomial plus exterior degree. This homotopy is coefficientwise finite, preserves each finite coefficient window, and therefore extends to the coefficient and length completions. Since the filtrations are complete and separated, the filtered map $\widehat{\Omega}\, C_*^{\mathrm{CE}}(L) \to \widehat{U}(L)$ is a quasi-isomorphism. Finally, Tate reflexivity identifies the continuous bar dual of $C^\bullet_{\mathrm{CE,cont}}(L)$ with this completed cobar algebra. □

Proposition 0.2.47 *(Koszul dual of continuous CE cochains).* Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ be the completed graded Lie algebra above, and let

$$B = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big).$$

Then

$$B^! \simeq \widehat{U(\mathfrak{g})},$$

the completed graded universal enveloping algebra.

Proof. Put

$$H_d = \mathrm{Span}\{z_1^a z_2^b : a + b = d\}, \qquad \mathfrak{h} = \prod_{d\geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} H_d^\vee.$$

Thus

$$\mathfrak{g} = \left(\prod_{d\geq 1} H_d\right) \ltimes \left(\bigoplus_{d\geq 1} H_d^\vee\right)[1]$$

is a Tate Lie algebra. The bracket is degreewise finite:

$$\{H_p, H_q\} \subset H_{p+q-2},$$

with the degree-zero summand projected to zero. Thus the degree-r coefficient of $\{f, g\}$ depends only on the finitely many pairs $p + q = r + 2$. For the coadjoint action, if $f \in H_p$ and $\lambda \in H_q^\vee$, then

$$\mathrm{ad}_f^\vee \lambda \in H^\vee_{q-p+2}$$

when $q - p + 2 \geq 1$, and it is zero otherwise. Finite-support Taylor duals remain finite-support: for fixed $\lambda \in H_q^\vee$ only the Hamiltonian degrees $p \leq q + 1$ can act nontrivially, and for a finite direct-sum support in $\mathfrak{h}^\vee$ the union of those contributing p is finite. Thus the product/direct-sum coefficient-finiteness hypotheses of Lemma 0.2.46 hold. The abelian bracket on $\mathfrak{h}^\vee[1]$ is trivial. The Taylor coefficient topology is independent of the PBW/order filtration used for the Rees algebra below.

Let

$$C_*^{\mathrm{CE}}(\mathfrak{g}) = \bigoplus_{n\geq 0} \widehat{\mathbf{S}}^n(\mathfrak{g}[1])$$

be the coefficient-completed, length-direct-sum Chevalley-Eilenberg coalgebra, with coderivation determined by the bracket of $\mathfrak{g}$. By definition

$$B = \mathrm{Hom}_{\mathrm{cont}}\big(C_*^{\mathrm{CE}}(\mathfrak{g}), \mathbb{C}\big).$$

The Chevalley-Eilenberg coalgebra is conilpotent for symmetric length. The augmentation $B \to \mathbb{C}$ has the continuous bar resolution in the category of complete filtered Tate spaces, and Lemma 0.2.46 identifies

$$\mathrm{RHom}_B(\mathbb{C}, \mathbb{C}) \simeq \widehat{\Omega}\, C_*^{\mathrm{CE}}(\mathfrak{g}),$$

the completed cobar algebra of the CE coalgebra. The completed Lie bar-cobar comparison in the lemma identifies this cobar algebra with the completed associative algebra generated by $\mathfrak{g}$ modulo the closed graded Lie relations; this is $\widehat{U(\mathfrak{g})}$.

Thus, for the generators $f, g \in \mathfrak{h}$ and $\eta_\lambda, \eta_\mu \in \mathfrak{h}^\vee[1]$, the completed enveloping algebra has relations

$$\begin{aligned} fg - gf &= \{f, g\}, \\ f\eta_\lambda - \eta_\lambda f &= \eta_{\mathrm{ad}_f^\vee \lambda}, \\ \eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda &= 0. \end{aligned}$$

□

For comparison with the classical brane algebra introduce the completed PBW/order Rees enveloping algebra

$$U_\hbar(\mathfrak{g}) = \widehat{T}(\mathfrak{g})[[\hbar]] \Big/ \overline{\langle xy - (-1)^{|x||y|} yx - \hbar[x, y] \rangle},$$

where the closure is for tensor length and the Tate coefficient topology. This Rees parameter records word length in the enveloping algebra and is independent of Taylor degree. At $\hbar = 0$ one obtains the completed PBW special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big).$$

After choosing the monomial-dual Taylor basis $\mathfrak{h}^\vee \cong \bar{A}_{\mathrm{desc}}$, the degreewise polynomial subalgebra $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^\vee[1])$ is the stable classical brane Hamiltonian descendant algebra of Proposition 0.2.40. Thus the degreewise polynomial part of the PBW associated graded of the closed Koszul dual is the stable classical brane Hamiltonian descendant algebra. The interacting closed Hamiltonian bracket deforms the closed special fibre to $\widehat{U(\mathfrak{g})}$; on the brane side the deformation checked here is the degree-zero Hamiltonian trace bracket below.

0.2.9.1 First-order corrections, order hbar one

Under the Koszul duality correspondence we have

$$z_1^k z_2^l \leftrightarrow \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The reduced Poisson bracket on the gravity side is:

$$\{z_1^k z_2^l,\, z_1^m z_2^n\}_{\bar{A}} = \overline{(kn - lm)\, z_1^{k+m-1} z_2^{l+n-1}}.$$

If an exponent in the displayed monomial is negative then the coefficient $kn - lm$ is already zero; if the displayed monomial is constant, its class in $\bar{A}$ is zero. The matching bracket on the open side is the canonical bracket induced by the first-order kinetic term $\int \mathrm{Tr}(\phi_1\, d\phi_2)$, with the sign convention $\iota_{X_f}\omega = -df$ fixed in the AKSZ remark above:

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j, \qquad \{(\phi_1)^i{}_j, (\phi_1)^k{}_l\} = 0 = \{(\phi_2)^i{}_j, (\phi_2)^k{}_l\}.$$

Lemma 0.2.48 *(Hamiltonian trace bracket).* The canonical matrix Poisson bracket descends to the invariant reduced BV cohomology and then to the Hamiltonian Lie quotient by constant trace classes.

Proof. For $\epsilon \in \mathfrak{gl}_N$, the component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ of the moment map generates the infinitesimal conjugation action, with the sign fixed by the convention for the bracket. Hence $\{F, \mu_\epsilon\} = 0$ for every invariant function F. If I_μ is the ideal generated by all μ_ϵ, then $\{F, \mu_\epsilon K\} = \mu_\epsilon\{F, K\}$ lies again in I_μ. Therefore the bracket of invariant functions descends to $(\mathcal{R}_N/I_\mu)^{GL_N}$, equivalently to the degree-zero reduced BV cohomology. The central line $\mathbb{C} \cdot \mathrm{Tr}(1)$ is central for this Lie bracket, so the Hamiltonian vector-space quotient by constants is compatible with the bracket. No associative quotient by the unit is taken. □

Proposition 0.2.49 *(First-order trace bracket).* In the stable connected Hamiltonian trace cohomology,

$$\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})}.$$

Proof. Write $F_{k,l} = \mathrm{Tr}(\phi_1^k \phi_2^l)$. The matrix Poisson bracket induced by

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$$

is

$$\{F, G\} = \mathrm{Tr}\left(\frac{\partial F}{\partial \phi_1}\frac{\partial G}{\partial \phi_2} - \frac{\partial F}{\partial \phi_2}\frac{\partial G}{\partial \phi_1}\right),$$

where the derivatives are cyclic matrix derivatives. Varying one occurrence at a time gives

$$\frac{\partial F_{k,l}}{\partial \phi_1} = \sum_{a=0}^{k-1} \phi_1^{k-1-a} \phi_2^l \phi_1^a,$$
$$\frac{\partial F_{k,l}}{\partial \phi_2} = \sum_{b=0}^{l-1} \phi_2^{l-1-b} \phi_1^k \phi_2^b.$$

Therefore the first summand in the bracket is a sum of kn traces, each obtained by removing one ϕ_1 from the first word and one ϕ_2 from the second word. The second summand is a sum of lm traces obtained by removing one ϕ_2 from the first word and one ϕ_1 from the second. Every trace appearing in either nonzero sum has multidegree $(k + m - 1, l + n - 1)$. Lemma 0.2.37 identifies all cyclic words with this multidegree in reduced BRST cohomology. The coefficient is therefore $kn - lm$, and the common class is $\overline{\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})}$. When the common word has multidegree $(0, 0)$, this class is the scalar trace and is zero in the Hamiltonian quotient by constants. □

Proposition 0.2.50 *(Open action on primitive descendants).* Let

$$\widetilde{\Psi}_{a,b} = \binom{a+b}{a}^{-1} \Psi_{a,b}$$

be the averaged primitive one-ψ class. In the stable primitive one-ψ homology,

$$\{\overline{\mathrm{Tr}(\phi_1^p \phi_2^q)}, \widetilde{\Psi}_{a,b}\} = (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1},$$

with the convention that a term with a negative exponent is zero. Thus the unreduced polynomial descendant classes carry the adjoint polynomial Hamiltonian action. The Hamiltonian descendant sector is the quotient by the scalar line $\mathbb{C}\widetilde{\Psi}_{0,0}$, so the displayed action is projected further by killing the scalar target.

Proof. Put $x = \phi_1$ and $y = \phi_2$. The cyclic derivatives of $F_{p,q} = \mathrm{Tr}(x^p y^q)$ are

$$\frac{\partial F_{p,q}}{\partial x} = \sum_{i=0}^{p-1} x^{p-1-i} y^q x^i, \qquad \frac{\partial F_{p,q}}{\partial y} = \sum_{j=0}^{q-1} y^{q-1-j} x^p y^j.$$

For a one-ψ cyclic word $\mathrm{Tr}(\psi w)$ of bidegree (a, b), differentiating with respect to y gives one summand for each of the b occurrences of y in w, and differentiating with respect to x gives one summand for each of the a occurrences of x. Multiplying by the displayed cyclic derivatives of $F_{p,q}$ therefore produces pb positive one-ψ words and qa negative one-ψ words for each $w \in \mathsf{W}_{a,b}$, all of total bidegree $(a + p - 1, b + q - 1)$.

The bracket is a cycle because $F_{p,q}$ is invariant, hence $\{F_{p,q}, \mu_\epsilon\} = 0$ for every moment-map component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[x, y])$. Equivalently, the Hamiltonian vector field of $F_{p,q}$ preserves the Koszul differential $Q\psi = [x, y]$ and acts on Koszul homology. By Proposition 0.2.34, the target primitive one-ψ homology in that bidegree is one-dimensional. The functional $\epsilon(\mathrm{Tr}(\psi w)) = 1$ detects its coefficient. Since $\epsilon(\widetilde{\Psi}_{a,b}) = 1$, the averaged input gives

$$\epsilon\big(\{F_{p,q}, \widetilde{\Psi}_{a,b}\}\big) = pb - qa.$$

Also $\epsilon(\widetilde{\Psi}_{a+p-1,b+q-1}) = 1$. Hence the displayed formula follows. If the target bidegree is $(0, 0)$, the same formula is still the unreduced primitive one-ψ action; the Hamiltonian descendant quotient sends that scalar class to zero. □

COROLLARY 0.2.51 *(Descendant action differs from the coadjoint action).* Under the boundary pairing $\delta_{a,b}[1] \mapsto [\Psi_{a,b}]$, the open Poisson action on polynomial one-ψ descendants differs from the closed coadjoint action on $\mathfrak{h}^\vee[1]$.

Proof. Proposition 0.2.50 gives

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\widetilde{\Psi}_{a+p-1,b+q-1}.$$

The closed coadjoint action is dual to the Poisson bracket. For $f = z_1^p z_2^q$ and the Taylor-dual basis vector $\delta_{a,b}$,

$$(\mathrm{ad}_f^\vee \delta_{a,b})(z_1^m z_2^n) = -\delta_{a,b}\big((pn - qm) z_1^{p+m-1} z_2^{q+n-1}\big).$$

Therefore, with the convention that negative indices give zero,

$$\mathrm{ad}^\vee_{z_1^p z_2^q} \delta_{a,b} = -(pb - qa + p - q)\delta_{a-p+1,b-q+1}.$$

For example, z_1 acts on open descendants by

$$\widetilde{\Psi}_{a,b} \longmapsto b\,\widetilde{\Psi}_{a,b-1},$$

whereas it acts coadjointly by

$$\delta_{a,b} \longmapsto -(b + 1)\delta_{a,b+1}.$$

The scalar edge $\{\,\mathrm{Tr}(\phi_1), \widetilde{\Psi}_{0,1}\,\} = \Psi_{0,0}$ is removed with the scalar descendant. The theorem uses Corollary 0.2.36 for the PBW special-fibre product identification. A descendant deformation match requires a different boundary realization of the continuous dual, such as a Laurent or distributional model, together with its own bracket calculation. □

Proposition 0.2.52 *(Principal-part realization of the coadjoint module).* Let

$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

with residue pairing

$$\langle \rho_{a,b}, z_1^m z_2^n\rangle = \mathrm{Res}_{z_1=z_2=0}\, z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}.$$

Let π_{pp} denote projection of Laurent two-forms to the span of the principal parts $\rho_{a,b}$. The action

$$f\cdot\rho = \pi_{\mathrm{pp}}\big(\{f,\rho\}\big)$$

identifies $\mathcal{P}$ with the continuous coadjoint module $\mathfrak{h}^\vee_{\mathrm{cont}}$. On monomials,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with the convention that a term with a negative index is zero.

Proof. The residue pairing identifies $\rho_{a,b}$ with the Taylor-dual functional $\delta_{a,b}$. Hamiltonian vector fields preserve the holomorphic volume form $dz_1 dz_2$, so integration by parts for the residue pairing gives

$$\langle \{f,\rho\}, g\rangle = -\langle \rho, \{f,g\}\rangle.$$

Hence the Laurent principal-part action realizes the coadjoint action. Directly, for $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$,

$$\begin{aligned}\{f, z_1^{-a-1} z_2^{-b-1}\} &= p z_1^{p-1} z_2^q\,(-(b+1)) z_1^{-a-1} z_2^{-b-2}\\ &\quad - q z_1^p z_2^{q-1}\,(-(a+1)) z_1^{-a-2} z_2^{-b-1}\\ &= -(pb - qa + p - q) z_1^{p-a-2} z_2^{q-b-2}.\end{aligned}$$

Since $z_1^{p-a-2} z_2^{q-b-2}\, dz_1 dz_2 = \rho_{a-p+1,b-q+1}$ when both indices are nonnegative, principal-part projection gives the displayed formula. □

Proposition 0.2.53 *(No polynomial boundary realization of the principal parts).* In the present polynomial open BV/Koszul matrix model, no family of ordinary polynomial boundary local cohomology classes

$$\Theta_{a,b}, \qquad a,b\geq 0, \quad a+b>0,$$

can realize the principal-part coadjoint module of Proposition 0.2.52.

Proof. The linear Hamiltonians

$$O_{1,0} = \overline{\mathrm{Tr}(\phi_1)}, \qquad O_{0,1} = \overline{\mathrm{Tr}(\phi_2)}$$

act on polynomial boundary observables by the constant Hamiltonian vector fields ∂_{ϕ_2} and $-\partial_{\phi_1}$, respectively. On every ordinary polynomial observable these operators are locally nilpotent: after enough iterations the polynomial degree in the differentiated variable is exhausted.

On the principal-part coadjoint module, however, Proposition 0.2.52 gives

$$O_{1,0} \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad O_{0,1} \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Neither operator is locally nilpotent on any nonzero principal-part generator. Local nilpotence is preserved under isomorphism of modules. Therefore the desired $\Theta_{a,b}$ cannot be ordinary polynomial boundary observables in the current open BV/Koszul model. □

Construction 0.2.54 *(Stalkwise boundary local-dual principal-part sector).* Let M_∂^{pp} be the reduced continuous dual of the stable connected Hamiltonian trace sector, with basis $\Theta_{a,b}$ dual to

$$O_{m,n} = \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}, \qquad m+n > 0,$$

by

$$\langle \Theta_{a,b}, O_{m,n} \rangle = \delta_{a,m}\delta_{b,n}.$$

Equivalently,

$$M_\partial^{\mathrm{pp}} \cong \mathrm{Ann}(\mathbb{C}\cdot 1) \subset H^2_{(z_1,z_2)}\big(\mathbb{C}[z_1,z_2]\big)\, dz_1 dz_2,$$

the continuous dual functionals vanishing on constants, with basis $\rho_{a,b}$ for $a+b>0$. Explicitly,

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a+b>0,$$

with

$$\Theta_{a,b} \longleftrightarrow \rho_{a,b}.$$

This is an invariant annihilator submodule, not a quotient by the non-invariant line $\mathbb{C}\,\rho_{0,0}$. Define the Hamiltonian action by negative transpose of the proved boundary Hamiltonian bracket:

$$\langle \{O_f, \Theta\}, O_g \rangle = -\langle \Theta, \{O_f, O_g\}_{\mathrm{open}} \rangle.$$

Then this stalkwise local-dual sector realizes the principal-part coadjoint action:

$$\{O_{p,q}, \Theta_{a,b}\}_{\mathrm{open}} = -(pb - qa + p - q)\Theta_{a-p+1,b-q+1},$$

with negative-index targets zero. Declare the odd local-dual ideal abelian:

$$\{\Theta_{a,b}, \Theta_{c,d}\}_{\mathrm{open}} = 0.$$

Equivalently, the assignment

$$z_1^p z_2^q \longmapsto O_{p,q}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

realizes stalkwise the Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$.

PROPOSITION 0.2.55 *(Reduced principal-part boundary current object).* After the de Rham contraction to the reduced local model on the brane line, the local-dual sector of Construction 0.2.54 extends to a compactly supported classical current object. For each interval $I \subset \mathbb{R}$, put

$$\mathfrak{J}_\partial^{\mathrm{pp}}(I) = \mathcal{D}'_c(I) \widehat{\otimes} \mathcal{P}[1],$$

where $\mathcal{D}'_c(I)$ denotes compactly supported distributions; regular compactly supported densities give the smooth smearing formulas below. Put also

$$\mathfrak{J}^{\mathrm{Ham}}_{\partial}(I) = C^{\infty}_c(I)\widehat{\otimes}\bar{A}$$

and define extension by zero for inclusions of intervals. Let

$$\mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}(I)\widehat{\otimes}\widehat{\mathbf{S}}\big(\mathfrak{J}^{\mathrm{pp}}_{\partial}(I)\big).$$

The differential is Q_{open} on the ordinary open factor and zero on $\mathfrak{J}^{\mathrm{pp}}_{\partial}(I)$. The product for disjoint intervals is the graded-commutative product after extension by zero.

For $a, b \in C^{\infty}_c(I)$, define the Hamiltonian smearings

$$B_f(a) = \int_I a(t)B_f(t)\,dt, \qquad O_f(a) = a \otimes f.$$

For $B \in \mathcal{D}'_c(I)$, write

$$\Theta_\rho(B) = B \otimes \rho[1].$$

Pair $\Theta_\rho(B)$ only with the primitive connected Hamiltonian trace projection by

$$\langle \Theta_\rho(B), B_g(b)\rangle = B(b)\rho(g).$$

Descendants, constants, and decomposable multi-traces are outside this pairing unless an additional coalgebra projection is specified. The P_0 bracket on the reduced current object is

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab), \qquad \{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\} = 0,$$

where aB is multiplication of a distribution by a smooth function. Thus, for regular densities $B = b(t)\,dt$ and on the basis $\Theta_{i,j}$,

$$\{O_{p,q}(a), \Theta_{i,j}(b)\} = -(pj - qi + p - q)\Theta_{i-p+1,\,j-q+1}(ab),$$

again with negative-index targets zero.

Proof. Extension by zero makes both $I \mapsto C^{\infty}_c(I)$ and $I \mapsto \mathcal{D}'_c(I)$ cosheaves on the line. Multiplication of a compactly supported distribution by a smooth compactly supported function is compatible with extension by zero. Tensoring with the fixed discrete principal-part module $\mathcal{P}[1]$ preserves this reduced compact-support functoriality. The factorization product on disjoint intervals is ordinary multiplication after pushing both compactly supported sections into the union; the odd generators Θ_ρ therefore generate the completed graded-symmetric, equivalently exterior, principal-part factor.

The differential is compatible with these structure maps because $Q_{\mathrm{open}}B_f = 0$ on the Hamiltonian trace classes and $Q\Theta_\rho = 0$ by definition in the added local-dual factor. The first bracket identity is Theorem 0.2.58 with test functions multiplied on the line. The second is the negative transpose of that bracket under the displayed primitive trace pairing, and reduces exactly to the principal-part coadjoint formula of Proposition 0.2.52. The third says that the shifted cotangent local-dual ideal is abelian. Jacobi is then the Jacobi identity in the semidirect Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$: equivalently, $[\mathrm{ad}^\vee_f, \mathrm{ad}^\vee_g] = \mathrm{ad}^\vee_{\{f,g\}}$ on the continuous dual. $\square$

Remark 0.2.56 *(Reduced status of the principal-part currents).* Proposition 0.2.55 is a reduced post-contraction P_0 current object. It is not an unreduced BV/factorization algebra. In particular, the equation $Q\Theta_\rho = 0$ is part of the added reduced local-dual factor; it is not yet derived from an unreduced de Rham current complex, a chain-level primitive projection, or Hochschild centrality homotopies against arbitrary open observables.

Problem 0.2.57 *(Factorization realization of the boundary local dual).* Proposition 0.2.55 supplies the post-contraction compact current object and the strict reduced bracket action for the principal-part modes. The remaining problem is to lift this reduced object to the unreduced boundary BV/factorization theory: define the compactly supported de Rham functoriality before contraction, the full BV differential and homotopies, the pairing with ordinary open observables before primitive trace projection, and the coherent P_0-center homotopies.

The expected unreduced enlargement has the form

$$\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I)\widehat{\otimes}\widehat{\mathbf{S}}\big(\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I)\big),$$

where

$$\mathcal{M}^{\mathrm{pp,unred}}_{\partial}(I) = C^{\mathrm{dR}}_{c,\mathrm{cur}}(I)\widehat{\otimes}\,\mathrm{Ann}(\mathbb{C}\cdot 1)[1].$$

Here $C^{\mathrm{dR}}_{c,\mathrm{cur}}(I)$ is the compactly supported de Rham current cosheaf on the line, with differential dual to $d_{\mathbb{R}}$, and $\mathrm{Ann}(\mathbb{C}\cdot 1)$ is the principal-part annihilator of constants. One must also construct a chain-level primitive, or cumulant, projection

$$\pi_{\mathrm{prim}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \longrightarrow \Omega^{\bullet}_{c}(I)\widehat{\otimes}\bar{A}$$

compatible with Q_{open}, constants, decomposable multi-traces, and factorization products. The required homotopies are the centrality homotopies for Θ_ρ against arbitrary unreduced open observables and the P_0 bracket homotopies

$$\{B_f, \Theta_\rho\} \simeq \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0$$

before primitive projection. Only after this enlargement can the full cotangent factorization-center lift use

$$\rho_{a,b}[1] \longmapsto \Theta_{a,b}.$$

THEOREM 0.2.58 *(Classical bulk-to-boundary Hamiltonian map).* In the stable connected Hamiltonian trace sector of the closed-open mixed holomorphic-topological theory, the boundary evaluation maps $\partial_{\mathrm{bb},N}$ of Construction 0.2.32 stabilize to a linear map

$$\partial_{\mathrm{bb}} : \bar{A} \longrightarrow H^0_{\mathrm{Ham}}(O^{\mathrm{cl}}_{\mathrm{brane}}), \qquad f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

This map is a morphism of Lie algebras from the closed Hamiltonian Poisson algebra $\bar{A}$ to the connected open Hamiltonian trace Poisson algebra:

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \partial_{\mathrm{bb}}\big(\{f, g\}_{\bar{A}}\big).$$

This is the checked degree-zero Hamiltonian bracket component expected from the closed-open map to the Poisson factorization center of the brane observables. The full Poisson factorization-center morphism is not constructed in this theorem.

Proof. The map is well-defined on $\bar{A}$ by Corollary 0.2.38; constants map to the scalar N and vanish in the Hamiltonian quotient by constants. Lemma 0.2.48 gives the open Hamiltonian trace Poisson bracket. For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, Proposition 0.2.49 gives

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

The closed Poisson bracket is

$$\{f, g\}_{\bar{A}} = \overline{(kn - lm)z_1^{k+m-1}z_2^{l+n-1}},$$

and applying ∂_{bb} gives the same expression. Bilinearity proves the identity for all polynomials. □

Corollary 0.2.59 *(Classical local bulk-boundary coupling).* For scalar compactly supported test functions a, b on the brane line, define the smeared boundary Hamiltonian

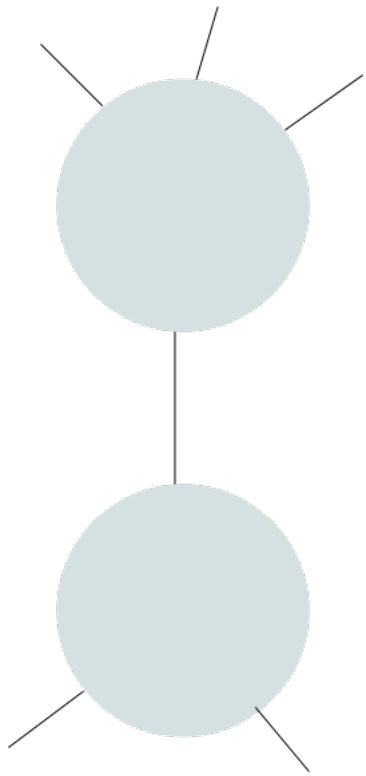
$$I_\partial(a, f) = \int_{\mathbb{R}} a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt$$

for $f \in \bar{A}$. After projection to the Hamiltonian quotient by scalar trace classes, and in the local equal-time Poisson bracket convention,

$$\{I_\partial(a, f), I_\partial(b, g)\}_{\mathrm{open}} = I_\partial(ab, \{f, g\}_{\bar{A}})$$

for $g \in \bar{A}$.

Proof. The test functions are scalar smearings of boundary observables. Form parts of the closed BV bracket are absent. The test functions have zero Poisson bracket with the matrix coordinates. The canonical bracket is local in t, so the equal-time delta-function multiplies the coefficients to $a(t)b(t)$, and the bracket of the inserted Hamiltonians is exactly the bracket computed in Theorem 0.2.58. Compact support removes boundary terms. □



This schematic picture records the single canonical contraction producing the coefficient $(kn - lm)$ in Proposition 0.2.49; it is not a separate Feynman-integral proof.

Proof of Theorem 0.1.17. Proposition 0.2.1 identifies the closed Hamiltonian BF sector and the compatible brane BV theory as local mixed holomorphic-topological data. The open Hamiltonian-sector algebra is obtained from the product of the topological self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$ and the formal holomorphic affine Koszul calculation for $\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m})$. Lemma 0.2.5 identifies the ghost-zero reduction with the gauged first-order matrix model

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Lemma 0.2.28 removes positive topological jets, and Lemma 0.2.37 identifies cyclic words with the same commutative monomial. Proposition 0.2.34 computes the primitive one-ψ homology, and Proposition 0.2.29 gives Proposition 0.2.40 after passing to the connected Hamiltonian trace sector:

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H} \cong \mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]).$$

On the closed side, Proposition 0.2.13 identifies the polynomial local Hamiltonian modes with $\bar{A}$, and their Taylor completion is $\mathfrak{h} = \widehat{\mathfrak{h}}$. The cotangent BF field supplies the shifted coadjoint copy, so the local Lie algebra is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$. Proposition 0.2.44 computes local observables as $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$. Proposition 0.2.47 gives the Koszul dual $\widehat{U(\mathfrak{g})}$, and the PBW/order Rees specialization gives $\widehat{\mathbf{S}}(\mathfrak{g})$ at

$\hbar = 0$. Its degreewise polynomial subalgebra is $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^\vee[1])$. Corollary 0.2.36 identifies $\mathfrak{h}^\vee$ with $\bar{A}_{\mathrm{desc}}$ as the nonconstant boundary-paired primitive one-ψ homology in the special fibre. Proposition 0.2.50 and Corollary 0.2.51 compute the open descendant bracket and separate it from the closed coadjoint deformation. Finally Theorem 0.2.58 constructs the bracket-compatible degree-zero Hamiltonian boundary evaluation from closed Hamiltonians to brane Hamiltonians. Corollary 0.2.59 packages this as the local classical bulk-boundary coupling. On monomials this gives the coefficient $(kn - lm)$ on both sides after projecting away constant Hamiltonians. □

0.2.9.2 Formal deformation-quantization target

The degree-zero Hamiltonian sector has a canonical formal deformation-quantization target. The formal holomorphic disk has constant Poisson tensor, so its canonical translation-invariant deformation is the Weyl/Moyal algebra. This fixes the algebraic coefficients one would have to recover from a quantum closed-open BV construction. It is not, by itself, a construction of that quantum BV theory. The degree-zero Hamiltonian bracket part of the classical theorem is the proposed special fibre of this formal deformation; the cotangent/descendant part remains the separate problem stated below. Put

$$A_\hbar = \big(\mathbb{C}[[z_1, z_2]][[\hbar]], *\big),$$

where $*$ is the Weyl/Moyal product of the next subsection. The symbol $\bar{A}_\hbar$ denotes the central quotient of the underlying Moyal Lie algebra by $\mathbb{C}[[\hbar]]$; it is not used as an associative quotient. We write

$$[f, g]_{\mathrm{raw}} = f * g - g * f, \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_{\mathrm{raw}}.$$

The degree-zero quantum Hamiltonian Lie algebra is

$$\mathfrak{h}_\hbar = \bar{A}_\hbar, \qquad [f, g]_{\mathfrak{h}_\hbar} = \{f, g\}_\hbar.$$

The specialization at $\hbar = 0$ is the Poisson Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ used above.

On the brane side, the open action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$$

makes the matrix entries of ϕ_1 and ϕ_2 conjugate variables. The canonical algebraic quantization of the unconstrained first-order matrix phase space is the completed Weyl algebra $\mathcal{W}_N$ generated by those entries with

$$[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\, \delta^i_l \delta^k_j, \qquad [(\phi_a)^i{}_j, (\phi_a)^k{}_l] = 0 \quad (a = 1, 2).$$

The gauge field imposes the conjugation moment-map constraint. With the commutator convention above, the normal-ordered quantum comoment is

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}\big([\epsilon, \phi_1]\phi_2\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

This formula is not cyclically reordered: for $\epsilon = 1$ it vanishes, as the scalar subgroup acts trivially. We use the convention

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a$$

for the induced infinitesimal conjugation action on the Weyl algebra; changing to the opposite commutator convention changes the displayed comoment by the corresponding sign. The quantum brane local algebra is the BRST quantum Hamiltonian reduction

$$\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = C^\bullet_{\mathrm{BRST}}\big(\mathfrak{gl}_N, \mathcal{W}_N, \widehat{\mu}\big),$$

whose degree-zero cohomology is the normalizer quotient

$$H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = \mathrm{N}_{\mathcal{W}_N}\big(\mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)\big)/\mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

The constant Hamiltonian trace is projected out in degree zero as before. The classical special fibre is the reduced BV algebra of Construction 0.2.27.

The quantum boundary map is Weyl-ordered trace evaluation:

$$\partial_{\mathrm{bb},\hbar,N}(f) = \overline{\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1,\phi_2)}, \qquad f \in \bar{A}_\hbar,$$

where Op_{W} denotes total symmetrisation before passing to the trace. At $\hbar = 0$ this recovers Construction 0.2.32. After dividing the commutator by $\hbar$ and reducing modulo $\hbar$, the raw commutator identity recovers Theorem 0.2.58.

PROPOSITION 0.2.60 *(BRST descent of Weyl-ordered Hamiltonian traces).* For every $f \in \bar{A}_\hbar$, the Weyl-ordered trace

$$F_{f,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1,\phi_2)$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F_{f,N}] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Hence $F_{f,N}$ lies in the normalizer of the quantum moment-map ideal and defines a degree-zero class in $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the scalar Hamiltonian.

Proof. Total Weyl symmetrisation is equivariant for linear changes of the matrix variables, and simultaneous conjugation acts linearly on (ϕ_1,ϕ_2). Therefore $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1,\phi_2)$ is invariant under infinitesimal conjugation. By the defining property of the quantum comoment,

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a,$$

this invariance is exactly $[\widehat{\mu}_\epsilon, F_{f,N}] = 0$. If $I = \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)$ is the left quantum moment-map ideal, then for every generator $\widehat{\mu}_\epsilon$, $\widehat{\mu}_\epsilon F_{f,N} = F_{f,N}\widehat{\mu}_\epsilon \in I$. Thus $F_{f,N}$ normalizes I and descends to the normalizer quotient describing $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$. □

Remark 0.2.61 *(Normal-ordering obstruction).* Proposition 0.2.60 is not the quantum Moyal theorem. Put $X = \phi_1$ and $Y = \phi_2$. With the normal ordering above,

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}([\epsilon, X]Y) = \mathrm{Tr}\big(\epsilon(YX - XY + \hbar N I)\big).$$

For $\epsilon = I$ this comoment is identically zero; equivalently the displayed matrix has trace zero inside the Weyl algebra. Thus the nontrivial quantum moment equations are represented by the traceless matrix relation

$$YX - XY + \hbar N I = 0,$$

not by the classical commuting-pair equation. Consequently, for a covariant matrix word A,

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\,\mathrm{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

This Capelli-type shift can feed lower-degree traces into the connected sector. A stable quantum theorem must prove that the chosen connected extraction removes or renormalizes this shift, or else change the sector or normalization.

Remark 0.2.62 *(Capelli-renormalized trace generators).* Let

$$T_{a,b} = \mathrm{Tr}(X^a Y^b)$$

denote the normal-ordered trace. The Weyl-ordered trace

$$J_N(z_1^a z_2^b) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(z_1^a z_2^b)(X, Y)$$

is triangular over the normal-ordered traces after imposing the quantum moment relation:

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}.$$

For example,

$$J_N(z_1 z_2) = T_{1,1} - \frac{\hbar N^2}{2}, \qquad J_N(z_1^2 z_2) = T_{2,1} - \hbar N\, T_{1,0},$$

and

$$J_N(z_1 z_2^2) = T_{1,2} - \hbar N\, T_{0,1}, \qquad J_N(z_1^2 z_2^2) = T_{2,2} - 2\hbar N\, T_{1,1} + \frac{\hbar^2 N^3}{2}.$$

The naive normal-ordered basis therefore has the wrong commutator even in low degree; a direct calculation gives

$$\hbar^{-1}[T_{2,1}, T_{0,2}] = 4T_{1,2} - 2\hbar N\, T_{0,1} \mod \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N),$$

whereas the scalar Moyal bracket would give only $4T_{1,2}$. Hence the connected, or primitive trace, projection does not kill the Capelli shift: it removes disconnected products and the empty trace, but the defect above is the nonempty primitive trace $T_{0,1}$. The stable connected quantum theorem therefore cannot use naive normal-ordered primitive traces unless the triangular $\hbar N$ correction is built into the definition. The natural finite-N generators are the Capelli-renormalized Weyl traces $J_N(f)$, equivalently

$$J_N(f) = T_N\left(\exp\left(-\frac{\hbar N}{2}\partial_{z_1}\partial_{z_2}\right) f\right)$$

in the normal-ordered trace notation $T_N(z_1^a z_2^b) = T_{a,b}$.

Theorem 0.2.63 *(Finite-N radial-parts Moyal identity).* Let $\mathcal{W}_N$ be the Weyl algebra of $T^*\mathrm{Mat}_N$ and let $\widehat{\mu}$ be the normal-ordered conjugation comoment above. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, put

$$J_N(f) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(X, Y).$$

Let $\mathfrak{t} \subset \mathfrak{gl}_N$ be the diagonal Cartan, with coordinates $\lambda_1, \ldots, \lambda_N$, and let

$$\Delta = \prod_{i<j}(\lambda_i - \lambda_j).$$

Put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Then the Harish-Chandra radial-part map, in the convention conjugated by the Vandermonde, satisfies

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f) \Delta.$$

Consequently, for all f, g the defect

$$D_N(f,g) = \hbar^{-1}[J_N(f), J_N(g)] - J_N(\{f,g\}_\hbar)$$

lies in the quantum moment-map ideal and therefore vanishes in the degree-zero quantum Hamiltonian reduction

$$\mathrm{N}_{\mathcal{W}_N}\big(\mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)\big)/\mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

Proof. Identify $\mathcal{W}_N$ with $\mathcal{D}_\hbar(\mathfrak{gl}_N)$, where X is multiplication and $Y = -\hbar\partial_X^t$. The quantum Hamiltonian reduction for conjugation is the normalizer quotient by the left ideal generated by the traceless comoment. The scalar matrix has zero comoment, as in Remark 0.2.61; equivalently the reduction is the $\mathfrak{sl}_N$ reduction together with GL_N-invariants. Since GL_N is reductive in characteristic zero, invariants are exact.

Harish-Chandra's radial-part construction, in the form refined by Wallach and by Levasseur–Stafford [9–12], gives an injective Harish-Chandra map from this quantum Hamiltonian reduction to the Weyl-invariant differential operators on the regular Cartan, after conjugation by Δ. It is therefore enough to compute the radial part on the generators $J_N(f)$.

Let $\mathcal{S}$ be the set of $f \in \mathbb{C}[z_1, z_2][[\hbar]]$ for which $\mathrm{Rad}_0(J_N(f)) = \Delta^{-1}S_N(f)\Delta$. Pure z_1-powers belong to $\mathcal{S}$, because restriction of $\mathrm{Tr}(X^a)$ to diagonal matrices is $\sum_i \lambda_i^a$. Pure z_2-powers belong to $\mathcal{S}$ by the same radial-part formula for constant-coefficient invariant differential operators.

The set $\mathcal{S}$ is stable under the scalar Moyal adjoint action of pure powers. Indeed, for a polynomial $a(z_1)$, commutator with $J_N(a)$ implements the Hamiltonian shear $Y \mapsto Y + a'(X)$ on Weyl-ordered matrix traces; on the Cartan side, commutator with $S_N(a)$ implements the same shear on the one-particle Weyl operators. The proof for a polynomial $b(z_2)$ is the dual shear. Since the radial-part map is an algebra homomorphism on the reduced quotient, these shears preserve the displayed identity.

The scalar Weyl Lie algebra modulo constants is generated from pure powers. For $a, b > 0$,

$$\{z_1^{a+1}, z_2^{b+1}\}_\hbar = (a+1)(b+1)z_1^a z_2^b + \text{terms of lower total degree.}$$

Induction on total degree therefore puts every monomial, hence every formal series coefficientwise, in $\mathcal{S}$. This proves the radial formula for all f.

Finally,

$$\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f,g\}_\hbar),$$

because the ith summand is the ordinary one-particle Weyl algebra and distinct summands commute. Conjugation by Δ preserves commutators, so $\mathrm{Rad}_0(D_N(f,g)) = 0$. Injectivity of the Harish-Chandra reduction map gives

$$D_N(f,g) \in \big(\mathcal{W}_N\widehat{\mu}(\mathfrak{sl}_N)\big)^{GL_N},$$

hence $D_N(f,g)$ is zero in the degree-zero quantum Hamiltonian reduction. □

Problem 0.2.64 *(Degree-zero quantum Hamiltonian-sector upgrade).* Construct the stable connected single-trace quotient of $\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ and prove that $\partial_{\mathrm{bb},\hbar,N}$ stabilizes to a map

$$\partial_{\mathrm{bb},\hbar} : \mathfrak{h}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$$

satisfying

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f,g\}_\hbar).$$

Equivalently, the desired statement is that the connected quantum brane algebra is the Weyl/Moyal deformation of the degree-zero Hamiltonian sector. The associated graded of this proposed degree-zero quantum boundary comparison at $\hbar = 0$ is the degree-zero part of Theorem 0.1.17. The finite-N radial-parts prerequisite is Theorem 0.2.63. Proposition 0.2.60 proves only the BRST descent of the individual Weyl-ordered boundary observables; it does not prove this stable connected commutator identity. The proof must in particular control the normal-ordering shift of Remarks 0.2.61 and 0.2.62: either the stable primitive generators are the Weyl/Capelli-renormalized classes $\overline{J_N(f)}$, or the connected-trace projection must explicitly subtract the triangular $\hbar N$ lower-trace terms.

Remark 0.2.65. The degree-zero Hamiltonian part has a canonical algebraic target: the Weyl/Moyal algebra of the formal symplectic disk, the constant-Poisson case of deformation quantization [8]. Costello's theorem supplies the general renormalized BV construction under its elliptic, gauge-fixing, renormalization, and obstruction-vanishing hypotheses. Costello–Li construct quantum BCOV theory and prove a specific flat open-closed BCOV/HCS quantization theorem [14, 15]. Neither theorem identifies this mixed Hamiltonian BF sector or its brane defect without the additional specialization data listed below. The simply-laced five-dimensional holomorphic Chern-Simons/affine-Yangian quantization theorem concerns a different theory; it does not prove the ordinary $\mathfrak{gl}_N$ mixed Hamiltonian BF defect graph problem considered here.

Remark 0.2.66. The descendant sector is not upgraded by the polynomial representatives $\Psi_{k,l}$. Corollary 0.2.51 shows that their open Hamiltonian action is the adjoint polynomial action, while the closed cotangent field carries the coadjoint action. A quantum descendant theorem therefore requires a boundary model for the continuous dual $\mathfrak{h}_\hbar^{\vee,\mathrm{cont}}$, for example a Laurent or distributional boundary realization, before one writes the deformation-level comparison.

Remark 0.2.67. The full cotangent deformation would replace $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ by

$$\mathfrak{g}_\hbar = \mathfrak{h}_\hbar \ltimes \mathfrak{h}_\hbar^{\vee,\mathrm{cont}}[1]$$

and would ask for $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_\hbar)$ as the closed quantum observable algebra. This is a separate descendant problem, not part of Problem 0.2.64, because the polynomial boundary representatives do not realize the coadjoint deformation-level action.

0.2.9.3 Imported graph package and specialization problem

The graph problem is the analytic form of the same statement. The algebra says which coefficients must appear; a closed-open BV theory must produce those coefficients as renormalized observable products, with propagator, counterterm, anomaly, and trace-normalization choices fixed. Costello's formalism supplies the general construction. The present paper must still identify the Hamiltonian input to that construction.

Statement 0.2.68 *(Imported Costello perturbative BV package).* For an elliptic BV theory with a gauge fixing, heat kernel K_t, propagator

$$P_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}} K_t \, dt,$$

and a renormalization scheme, Costello constructs finite-scale effective interactions by the RG operator

$$W(P_{\epsilon,L}, I[\epsilon]) = \hbar \log\bigl(\exp(\hbar \partial_{P_{\epsilon,L}}) \exp(I[\epsilon]/\hbar)\bigr).$$

The renormalized interaction is obtained by adding local counterterms and taking the $\epsilon \to 0$ limit. It has the connected-graph expansion

$$I[L] = \sum_{\Gamma \text{ connected}} \frac{\hbar^{b_1(\Gamma)+\sum_v g_v}}{|\operatorname{Aut}\Gamma|} W_\Gamma(P_{\epsilon,L}; I_{g_v}[\epsilon] + I^{\mathrm{CT}}_{g_v}(\epsilon)),$$

satisfies RG flow and locality, and satisfies the quantum master equation when the obstruction classes vanish [13]. The resulting quantum observables form a factorization algebra [16, 17].

Statement 0.2.69 *(Imported Costello–Li flat open-closed BCOV theorem).* Costello–Li prove a unique perturbative quantization of open-closed BCOV theory on flat $\mathbb{C}^d$, for d odd, with open gauge algebra $\mathfrak{gl}(N \mid N)$ and compatibility under $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N+k \mid N+k)$. In that flat open-closed BCOV/HCS theory they compute the anomaly cancellation needed for the graph construction [15]. Ordinary $\mathfrak{gl}_N$ and the mixed $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian brane model are not covered by this theorem without an additional reduction or summand argument.

Remark 0.2.70. The two imported statements supply the universal renormalized BV framework and one major flat BCOV/HCS instance, subject to their hypotheses. They do not by themselves prove the graph realization for the Hamiltonian sector of this paper. Here the brane $\mathbb{R} \times p$ is a defect in the mixed space, not a spacetime boundary unless a half-space model is separately chosen.

LEMMA 0.2.71 *(Finite-window obstruction for the full Hamiltonian sector).* The polynomial Hamiltonian Lie algebra of the formal symplectic plane admits no naive finite-dimensional Lie truncation compatible with the full sector used in this paper. More precisely:

(i) finite degree windows are not Lie subalgebras;
(ii) projected degree-window brackets need not satisfy Jacobi;
(iii) $\mathfrak{m}$-adic jet quotients are not Lie quotients in the presence of linear Hamiltonians;
(iv) there is no directed exhaustion by finite-dimensional Lie subalgebras containing the full polynomial Hamiltonian algebra.

Proof. Let H_d be the span of homogeneous polynomials of degree d, and let $\mathfrak{m} = (z_1, z_2)$. The Poisson bracket satisfies $\{H_p, H_q\} \subset H_{p+q-2}$. Thus

$$\{z_1^N, z_2^N\} = N^2 z_1^{N-1} z_2^{N-1},$$

which has degree $2N - 2 > N$ for $N > 2$, so degree $\leq N$ windows are not Lie subalgebras.

If one instead projects the bracket back to a degree window, Jacobi can fail. For the bracket $\pi_{\leq 3}\{-,-\}$, take

$$f = z_2, \qquad g = z_2^3, \qquad h = z_1^2 z_2.$$

A direct calculation gives the projected Jacobiator

$$[[f,g]_3, h]_3 + [[g,h]_3, f]_3 + [[h,f]_3, g]_3 = 6z_2^3.$$

The $\mathfrak{m}$-adic quotients are not stable under translations generated by linear Hamiltonians:

$$\{z_1^{N+1}, z_2\} = (N+1) z_1^N \notin \mathfrak{m}^{N+1}.$$

Hence $\mathfrak{h}/\mathfrak{m}^{N+1}$ is not a Lie quotient for the full Hamiltonian algebra.

Finally, any finite-dimensional Lie subalgebra containing

$$a = z_1^2 z_2^2, \qquad b = z_1^3 z_2$$

contains all iterated brackets

$$(\mathrm{ad}_a)^n(b) = (-4)^n z_1^{n+3} z_2^{n+1}, \qquad n \geq 0,$$

an infinite linearly independent set. Thus no directed finite-dimensional Lie-subalgebra exhaustion can contain all polynomial Hamiltonians. □

Remark 0.2.72 *(Linear heat operator versus BV kernel).* On the linear field complex

$$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

the flat gauge fixing

$$Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar\partial^*_{\mathbb{C}^2}, \qquad \Delta = [d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}, Q^{\mathrm{GF}}]$$

gives finite-scale heat operators

$$K_t = K_t^{\mathrm{base}}\widehat{\otimes}\mathrm{id}, \qquad P^{\mathrm{op}}_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}} K_t^{\mathrm{base}}\, dt\widehat{\otimes}\mathrm{id}$$

satisfying

$$[d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}, P^{\mathrm{op}}_{\epsilon,L}] = K_\epsilon - K_L$$

as continuous operators on the completed coefficient spaces. This is only an operator-level homotopy. To obtain Costello's BV propagator one also needs a coefficient coevaluation kernel for the Tate pair $\widehat{\mathfrak{h}} \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$. In the product/direct-sum topology used here, the formal sum of homogeneous coefficient coevaluations is not a convergent element of the completed tensor product. Thus the analytic graph problem requires a genuine Tate or weighted coefficient extension of the heat-kernel calculus, or a declared restriction to a smaller sector.

Lemma 0.2.73 *(Tate Casimir obstruction for the full Hamiltonian sector).* In the product/direct-sum Tate topology

$$\mathfrak{h} = \prod_{d\geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} H_d^\vee,$$

the coefficient coevaluation

$$C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_i e_{d,i} \otimes e_d^i$$

is not an element of $\mathfrak{h}\widehat{\otimes}\mathfrak{h}^\vee_{\mathrm{cont}}$, nor of $\mathfrak{h}^\vee_{\mathrm{cont}}\widehat{\otimes}\mathfrak{h}$. Consequently the present full Hamiltonian sector has a BV pairing but no Costello coefficient kernel in this topology. The operator-level homotopy of Remark 0.2.72 therefore does not define the BV propagator, the BV Laplacian, or renormalized graph weights for the full sector.

Proof. For each d, choose a basis $\{e_{d,i}\}_i$ of H_d and the dual basis $\{e_d^i\}_i$ of $H_d^\vee$. An element of a completed tensor product with the direct-sum factor $\bigoplus_d H_d^\vee$ has finite support in that direct-sum direction in the present Tate convention. The displayed formal Casimir has a nonzero component in every $H_d^\vee$. Hence it is not a convergent tensor. The same support obstruction holds with the tensor factors reversed. Costello's heat-kernel BV formalism requires the inverse pairing as an actual distributional kernel, not merely an abstract continuous operator on the field complex. □

Problem 0.2.74 *(Hamiltonian specialization of Costello perturbative BV).* Costello supplies the universal renormalized BV graph calculus. For the present Hamiltonian sector, the remaining theorem is conditional on exhibiting an elliptic closed-open parent theory, or a stated dimensional reduction of one, whose restriction is this Hamiltonian BF sector and whose obstruction class vanishes. The required data are:

(i) choose the route: either prove that the Hamiltonian cotangent BF complex

$$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big)$$

is literally a closed sector of a Costello-type open-closed BV theory, or prove that it is obtained by a specified dimensional reduction. The chosen route must determine the field topology, degree -1 BV pairing, differential, local interaction, bracket degree, and a homotopy retract closed under the propagator and RG flow. Since $\widehat{\mathfrak{h}}$ is a completed Tate coefficient system, the proof cannot use a naive finite-degree or $\mathfrak{m}$-adic truncation for the full sector, by Lemma 0.2.71. It must instead construct a genuine Tate or weighted-coefficient extension of Costello's heat-kernel calculus compatible with the Poisson bracket and RG flow, or restrict to a smaller Lie sector such as $\mathfrak{m}^2/\mathfrak{m}^{N+1}$ and explicitly declare that translations and the stated Moyal normalization have been removed. By Lemma 0.2.73, a damping factor inside the present direct-sum dual is not enough; the coefficient category itself must be replaced by one in which the Casimir converges, or the sector must be restricted;

(ii) specify the asymptotic conditions on $\mathbb{C}^2$ and the brane-defect condition on $\mathbb{R} \times p$;

(iii) choose the induced finite-scale gauge fixing, heat-kernel propagator, and BV Laplacian in this sector. The infrared limit $L \to \infty$ may be used only after the zero modes and harmonic projection have been fixed;

(iv) compute the obstruction/anomaly class for this mixed Hamiltonian sector. Concretely, the obstruction should be represented as a local functional class

$$\mathrm{Ob}_n \in H^1\big(\mathrm{Loc}_{\text{bulk+defect}}, Q + \{I_0, -\}\big)$$

coming from the n-loop failure of the quantum master equation after counterterms. Costello–Li's flat $\mathfrak{gl}(N \mid N)$ anomaly cancellation is a source theorem and a model, not an automatic transfer. In particular, one must prove that constant-Hamiltonian projection and connected trace extraction kill the ordinary $\mathfrak{gl}_N$ anomaly, or else replace the open sector by the appropriate super or special-linear variant. The first defect-local obstruction visible in the present algebra is the Capelli contact term

$$\omega_{\mathrm{Cap}}(A) = N\,\overline{\mathrm{Tr}(A)},$$

represented by

$$\mathrm{Tr}(A\,XY) - \mathrm{Tr}(A\,YX) = \hbar N\,\mathrm{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

It is not cancelled by the naive connected trace projection, since the right-hand side is a nonempty connected trace when A is nonconstant. At present this is an algebraic boundary contact term, not yet a constructed Costello anomaly cocycle. A QME proof must lift it to a degree-one local class and then either cancel this class by an explicit closed exchange graph and counterterm,

$$\omega_{\mathrm{Cap}} + W_{\text{closed exch}} + (Q + \{I_0, -\})C = 0,$$

remove it by a specified supertrace or central-character normalization, or build it into the renormalized boundary generators;

(v) construct the bulk-to-defect kernel as a cochain map to the Poisson factorization center of the brane observables. On degree-zero Hamiltonians it must send $f(z_1, z_2)$ to the Weyl-ordered boundary observable $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$, and it must satisfy the Ward identity and the quantum moment-map constraint;

(vi) define the relevant boundary ribbon graphs: cyclic order, boundary components, automorphism factors, propagator labels, and the relation between Costello's loop-counting $\hbar$ and the Weyl contraction parameter. Equivalently, one must prove that a single boundary cross-contraction is $P/2$ in the Weyl convention;

(vii) fix the one-propagator boundary graph normalization. For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the raw first boundary commutator must have coefficient

$$\hbar(kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

equivalently the normalized commutator must have coefficient

$$(kn - lm) z_1^{k+m-1} z_2^{l+n-1};$$

(viii) after the first normalization, compute the three-propagator boundary graph normalization and prove that the connected order-$\hbar^3$ commutator contributes exactly

$$\frac{\hbar^3}{24} P^3(f, g)$$

with $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$;

(ix) give the convention for extracting connected single-trace and multi-trace coefficients as cumulants or primitive trace classes, and prove that this extraction cancels, renormalizes, or otherwise accounts for the normal-ordering shift of Remark 0.2.61.

The computations here settle the formal polynomial Hamiltonian algebra and the formal Moyal coefficients. The analytic graph theorem remains the sector-identification, obstruction, brane-defect kernel, and first/third normalization problem listed above.

Remark 0.2.75 *(Convention firewall).* This local Hamiltonian BF sector is not a compact Calabi-Yau output of a Φ_d functor, carries no κ_{ch} or κ_{BKM} consequence, and is not evidence for compact CY_3 BCOV anomaly cancellation. The Moyal computation below is local deformation quantization of a formal symplectic disk. A closed-open factorization source is extra structure: descent, QME/anomaly cancellation, completion, and the specialization or defect map must be supplied before one may pass from a formal coefficient calculation to a Costello graph realization. The Weyl algebra fixes the target coefficients; it does not reconstruct the elliptic BV source. In particular the formal disk carries no compact Calabi-Yau datum, no $K3 \times E$ specialization, no chosen (Σ, C) specialization, no Igusa cusp form, no Borcherds input form, no BKM denominator identity, and no Stage-1 or Stage-2 Φ_d specialization.

0.2.9.4 Formal Weyl/Moyal target, all orders

We use the Weyl/Moyal Rees convention for the formal quantization of the constant Poisson tensor [8]:

$$f * g = m \circ \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g).$$

The factor $1/2$ is already forced by linear Hamiltonians: if $u = az_1 + bz_2$ and $v = cz_1 + dz_2$, then $[\widehat{u}, \widehat{v}] = \hbar(ad - bc)$, so the Baker-Campbell-Hausdorff formula for Weyl-ordered exponentials contributes

$\exp(\hbar P(u,v)/2)$. Differentiating these exponentials gives the cross-contraction $P/2$ in the formal Weyl convention. Then the star commutator is odd in $\hbar$:

$$[f,g]_* = \hbar\{f,g\} + \frac{\hbar^3}{24}P^3(f,g) + O(\hbar^5),$$

where $P = \partial_{z_1}\otimes\partial_{z_2} - \partial_{z_2}\otimes\partial_{z_1}$. Thus the $\hbar^2$ coefficient vanishes in this quantization convention. The normalized quantum Hamiltonian bracket is therefore

$$\{f,g\}_\hbar = \hbar^{-1}[f,g]_* = \sum_{s\geq 0}\frac{\hbar^{2s}}{2^{2s}(2s+1)!}\,P^{2s+1}(f,g).$$

PROPOSITION 0.2.76 *(Moyal coefficients on monomials).* Let $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$. For every $r \geq 0$,

$$P^r(f,g) = \sum_{a=0}^{r}(-1)^a\binom{r}{a}(k)_{r-a}(l)_a(m)_a(n)_{r-a}z_1^{k+m-r}z_2^{l+n-r},$$

where $(s)_q = s(s-1)\cdots(s-q+1)$ and $(s)_0 = 1$. Terms with a negative displayed exponent are understood to be zero; equivalently the corresponding falling factorial already vanishes. Consequently

$$[f,g]_* = \sum_{\substack{r\geq 1\\ r\text{ odd}}}\frac{2}{2^r r!}\,\hbar^r P^r(f,g),$$

so the second-order coefficient vanishes. The first four odd coefficients are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24}P^3(f,g), \qquad \frac{\hbar^5}{1920}P^5(f,g), \qquad \frac{\hbar^7}{322560}P^7(f,g).$$

Proof. Since

$$P = \partial_{z_1}\otimes\partial_{z_2} - \partial_{z_2}\otimes\partial_{z_1},$$

the two summands commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r}(-1)^a\binom{r}{a}\partial_{z_1}^{r-a}\partial_{z_2}^{a}\otimes\partial_{z_1}^{a}\partial_{z_2}^{r-a}.$$

Applying this to $z_1^k z_2^l \otimes z_1^m z_2^n$ gives the displayed falling factorial formula. Interchanging f and g changes $P^r(f,g)$ by the sign $(-1)^r$, so the star commutator keeps exactly odd r and contributes the factor $2/(2^r r!)$ in order r. □

PROPOSITION 0.2.77 *(Conditional all-order boundary-product normalization).* Suppose a Hamiltonian specialization of the Costello closed-open observable product has the following degree-zero boundary restriction: for Weyl-ordered Hamiltonian boundary observables, the product is

$$m\circ\exp\left(\frac{\hbar}{2}P_\partial\right), \qquad P_\partial = P,$$

and connected single-trace extraction introduces no additional factor of N, sign, or scalar moment-map term. Then the antisymmetrised boundary product has all odd raw commutator coefficients

$$\frac{2}{2^r r!}\hbar^r P^r(f,g), \qquad r\geq 1 \text{ odd}.$$

In particular, the first and third coefficients are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24}P^3(f,g).$$

Proof. This is Proposition 0.2.76 applied to the assumed boundary product. The exponential contributes $1/r!$, Weyl normalization contributes 2^{-r}, and antisymmetrisation keeps exactly the odd powers of P and adds the factor 2. Thus the coefficient in odd order r is $2/(2^r r!)$. For $r = 1$ this gives 1, and for $r = 3$ it gives $2/(2^3 3!) = 1/24$. □

Proposition 0.2.78 *(Open-line midpoint graph weights).* In the reduced open first-order line model

$$S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2),$$

choose the zero-mode-removed midpoint propagator

$$G(t,s) = \frac{1}{2}\,\mathrm{sgn}(t-s), \qquad \langle \phi_1(t)\phi_2(s)\rangle = \hbar G(t,s), \quad \langle \phi_2(t)\phi_1(s)\rangle = -\hbar G(t,s).$$

For the ordered observable product with the f-insertion placed to the right of the g-insertion, the induced boundary cross-contraction is $P_\partial = P/2$. Hence the one-edge ordered graph has weight $\hbar P(f,g)/2$, and the three-parallel-edge ordered graph has weight $\hbar^3 P^3(f,g)/48$. After subtracting the reverse ordered product, the commutator contributions are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24} P^3(f,g).$$

Proof. The midpoint value $G(+,-) = 1/2$ assigns one copy of

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$

to each cross-contraction, with the Weyl factor $1/2$. For one edge there is no nontrivial edge automorphism, so the ordered product contributes $\hbar P(f,g)/2$. The reverse ordered product contributes $\hbar P(g,f)/2 = -\hbar P(f,g)/2$, and the commutator gives $\hbar P(f,g)$.

For three parallel edges between the same two labelled observable vertices, the automorphism group permutes the three identical edges and has order $3!$. The ordered product contribution is therefore

$$\hbar^3 \frac{1}{3!}\left(\frac{1}{2}\right)^3 P^3(f,g) = \frac{\hbar^3}{48} P^3(f,g).$$

Since $P^3(g,f) = -P^3(f,g)$, subtracting the reverse ordered product doubles this coefficient and gives $\hbar^3 P^3(f,g)/24$. □

Remark 0.2.79. Proposition 0.2.78 is a computation in the reduced open-line Weyl model. It is not the Costello closed-open graph theorem. The latter still requires an elliptic parent or dimensional reduction, a brane-defect condition, finite-scale gauge fixing, the induced bulk-to-defect propagator, orientations and measures on configuration spaces, counterterms, anomaly cancellation, and compatibility with the quantum Hamiltonian reduction.

0.2.9.5 Third-order target for Costello's graph problem

For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the third-order Moyal coefficient from Proposition 0.2.76 is

$$P^3(f,g) = \sum_{a=0}^{3} (-1)^a \binom{3}{a} (k)_{3-a} (l)_a (m)_a (n)_{3-a} z_1^{k+m-3} z_2^{l+n-3},$$

where $(r)_s = r(r-1)\cdots(r-s+1)$ and $(r)_0 = 1$. Therefore

$$[f,g]_* = \hbar(kn-lm)z_1^{k+m-1}z_2^{l+n-1} + \frac{\hbar^3}{24}P^3(f,g) + O(\hbar^5).$$

The script `scripts/check_moyal_coefficients.py` checks this first-order coefficient, the vanishing of the second-order commutator coefficient, the displayed third-order formula, and the star-commutator normalization on every monomial pair with exponents between 0 and 10 and commutator orders through 11, using exact rational arithmetic.

This algebraic computation is an observable-product target, not an effective action computation. In Costello's formalism the connected effective action and the product/OPE of quantum observables are related but distinct. For two boundary observables F, G, the finite-scale free-field model has the schematic product

$$F \star_L G = m \circ \exp(\hbar\partial_{P_{\partial,L}})(F \otimes G),$$

where $P_{\partial,L}$ is the boundary or brane-defect propagator induced by the chosen kernel. The connected single-trace part is the cumulant, or primitive trace, extracted from this product. In this convention the order in $\hbar$ is the number of boundary cross-contractions; it is not the loop order of the connected effective interaction.

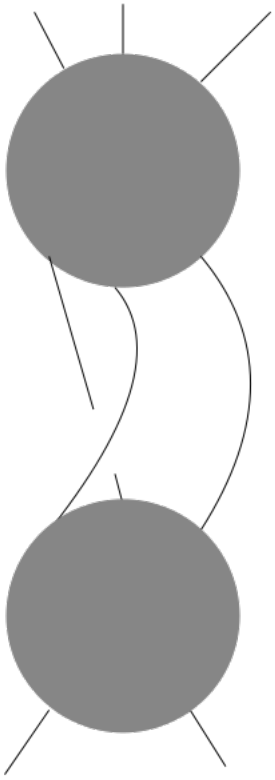
Thus there are two equivalent normalizations to keep separate. In Costello's observable-product notation one asks for the boundary contraction kernel

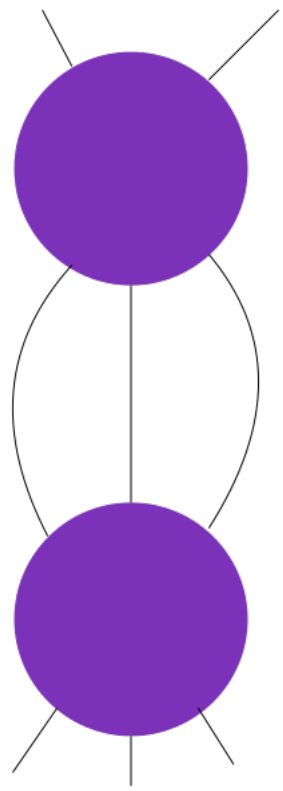
$$P_{\partial,L} \rightsquigarrow \frac{1}{2}P$$

on degree-zero Hamiltonians, so that

$$m \circ \exp(\hbar\partial_{P_{\partial,L}}) \rightsquigarrow m \circ \exp\left(\frac{\hbar}{2}P\right),$$

the Weyl/Moyal product used above. The ordered product has labeled inputs (f,g); its r identical cross-contractions contribute the denominator $r!$. No additional division by interchanging the two boundary vertices is made in the ordered product. The reverse order is the separate product $g * f$ whose subtraction forms the commutator.





These schematic pictures record the displayed order-$\hbar^3$ boundary contraction types whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the constant Poisson bivector P. Their analytic graph weights, automorphism factors, and counterterm cancellations are precisely the open checks in the following problem; the proved statement here is the algebraic Moyal coefficient.

Problem 0.2.80 *(First and third graph normalizations inside Costello theory).* Work inside the observable product supplied by Statement 0.2.68, after solving the Hamiltonian specialization data of Problem 0.2.74. The first graph fixes the normalization; the third graph tests whether the Costello specialization really lands on the Weyl/Moyal product rather than on another deformation quantization convention. The analytic problem is reduced to the following checks:

(i) identify the connected one-propagator brane-defect ribbon graph whose tensor contraction is $P(f, g)$, and prove that its antisymmetrised boundary product gives the raw coefficient $\hbar P(f, g)$;

(ii) fix the sign, automorphism factor, boundary orientation, and trace normalization in this first-order graph so that the coefficient is $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ on monomials;

(iii) identify the connected three-propagator brane-defect ribbon graphs whose tensor contraction is $P^3(f, g)$;

(iv) prove that every other connected order-$\hbar^3$ contribution in the Hamiltonian sector vanishes, is counterterm-exact, or lands in the discarded scalar trace;

(v) compute the graph weight and automorphism factor after the first-order normalization and show that the coefficient is $\frac{1}{24}$;

(vi) check that connected large-N extraction introduces no additional factor of N, sign, or scalar moment-map term.

The desired output is the pair of algebraic coefficients

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24} P^3(f,g)$$

in the Weyl/Moyal commutator. The formal coefficients are proved above; the analytic work is exactly the specialization and normalization check listed here.

Remark 0.2.81. In characteristic zero, GL_N is linearly reductive on rational representations. Therefore taking GL_N-invariants is exact, and

$$H\big((\mathcal{R}_N)^{GL_N}, Q\big) \cong H(\mathcal{R}_N, Q)^{GL_N}.$$

This justifies the invariant calculation used above. A Chevalley BRST model for the quotient stack introduces Lie algebra ghosts as presentation data. Linearly reductive exactness identifies derived algebraic invariants with ordinary invariants in this setting.

0.2.10 Further directions

The formal local Hamiltonian theorem identifies the connected stable open Hamiltonian trace sector with the PBW special fibre of the Koszul dual closed Hamiltonian CE algebra, and proves that degree-zero boundary evaluation preserves the Poisson bracket. The next closed-open step is the cochain-level Poisson factorization-center lift of Problem 0.2.25.

The quantum Weyl/Moyal formulas give the algebraic target for deformation, and Theorem 0.2.63 proves the finite-N reduced radial-parts identity for Weyl/Capelli generators. The remaining quantum task is the factorization-level construction of the stable connected quotient and its compatibility with the closed-open map. The analytic task is to identify the Costello graph package with this mixed Hamiltonian BF sector, including the brane-defect kernel, anomaly comparison, connected trace extraction, and the first/third graph normalizations of Problem 0.2.74. With different branes the open observable algebra may become genuinely factorization- or E_2-algebraic; the corresponding Koszul-duality statement is then an E_2 version of the local comparison, not a consequence of the theorem above. The same caveat applies to the expected D_0-brane analogue on $\mathbb{R}^2 \times \mathbb{C}^4$.

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